



AI in Drug Discovery – An Overview

Session 1

September 2, 2026

Who we are!

Pat Walters
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What we will do today

Session 0 – 1:00– 1:30 pm

- Introduction to Jupyter notebooks

Session 1 – 1:30 – 2:30 pm

- An introduction to Artificial Intelligence (AI) and Machine Learning (ML)
- Molecular representations
- AI architectures

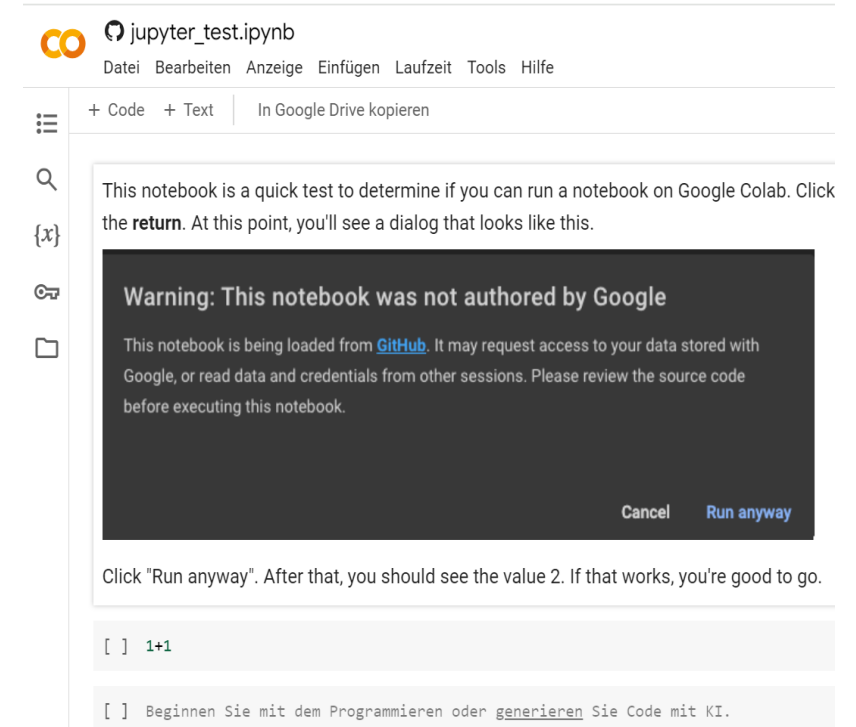
Session 2 – 3:00 – 4:00 pm

- Importance of data quality for AI and ML
- Data sets and (pre)processing
- Trusting your predictions

Session 3 – 4:30 – 5:30 pm

- Practical applications
- Generative AI
- Active learning

Lectures supported by hands-on sessions ...



Tutorials and Slides Can Be Found On GitHub

https://github.com/volkamerlab/ai_in_chemistry_workshop_2026

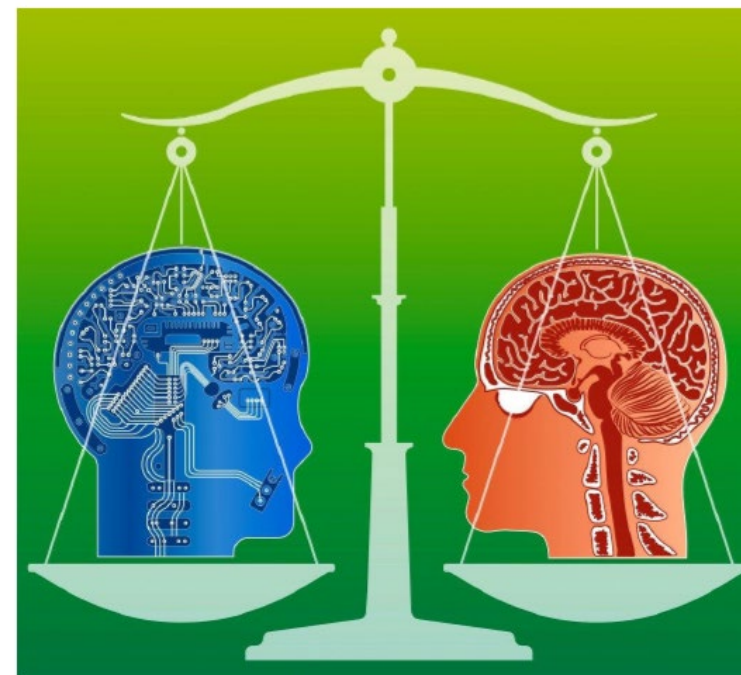
Human vs. Artificial Intelligence

Computer

- Fast and error-free information processing
- Processing of very large data volumes
- High-capacity memory
- Perpetual memory
- Very specialized ...

Human

- Good pattern recognition (based on a few examples)
- High generalizability
- Understanding, planning
- Empathy, awareness
- Flexibility
- Creativity ...

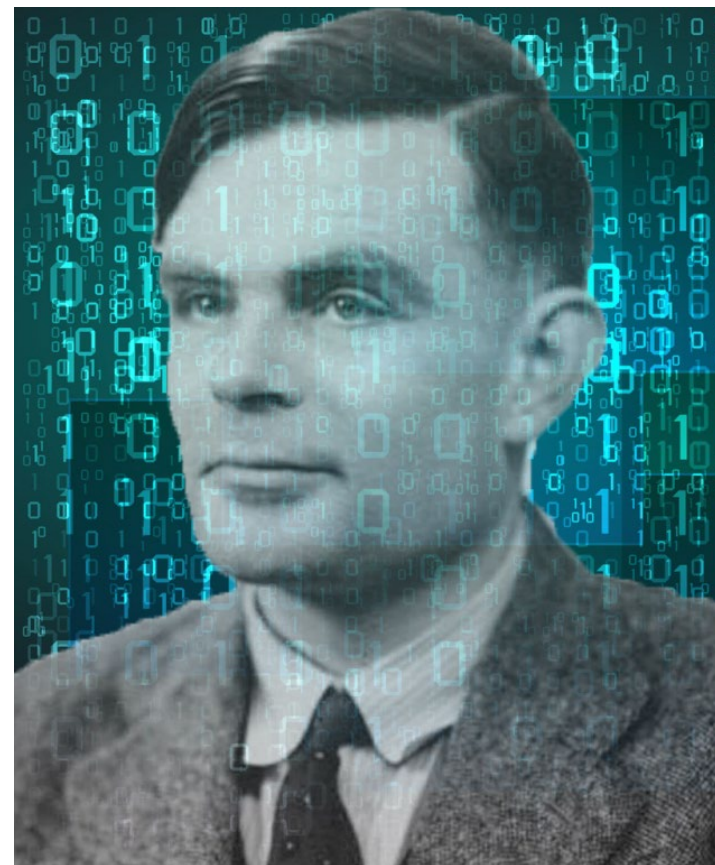
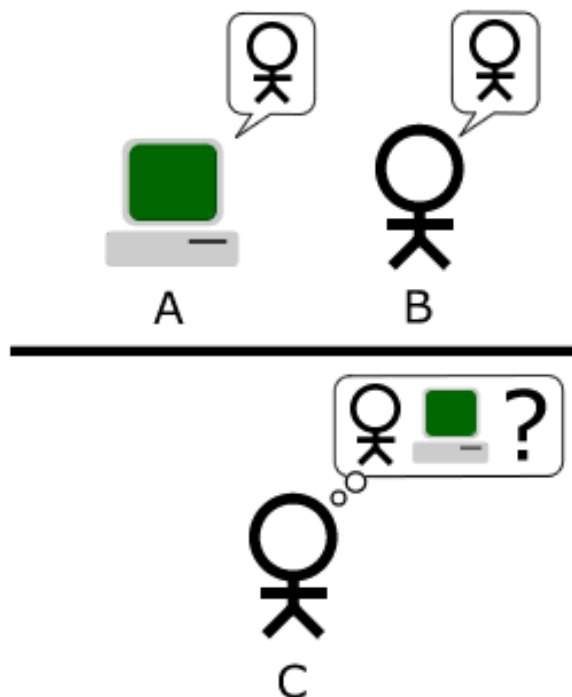


<https://www.scienceabc.com>

Artificial Intelligence and Machine Learning

1950

'Can machines think?'



Alan Turing, 1912–1954

Turing Test: <https://de.wikipedia.org/wiki/Turing-Test> Alan Turing: <https://www.history.org.uk>

Artificial Intelligence, Machine and Deep Learning



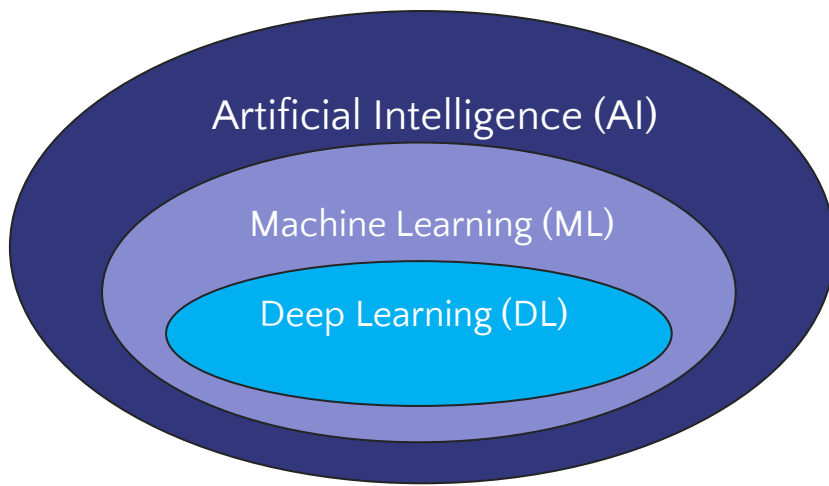
You

how would you define AI in simple words



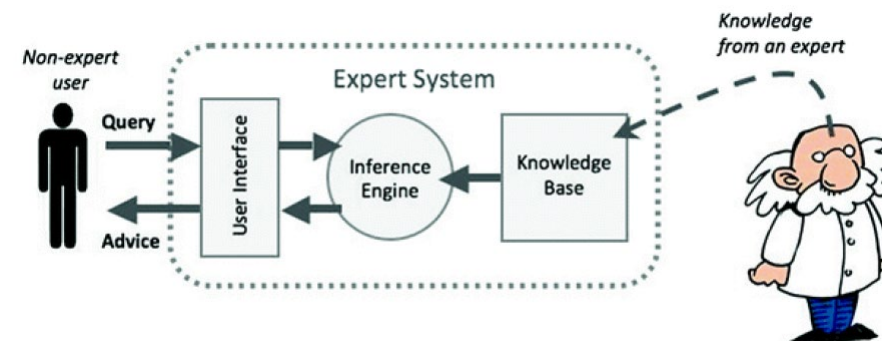
ChatGPT

Artificial Intelligence (AI) refers to the development of computer systems that can perform tasks that typically require human intelligence. These tasks include things like learning, reasoning, problem-solving, perception, and language understanding. AI aims to create machines that can mimic or simulate human cognitive functions, enabling them to adapt and improve their performance over time.



AI: Rule-based learning

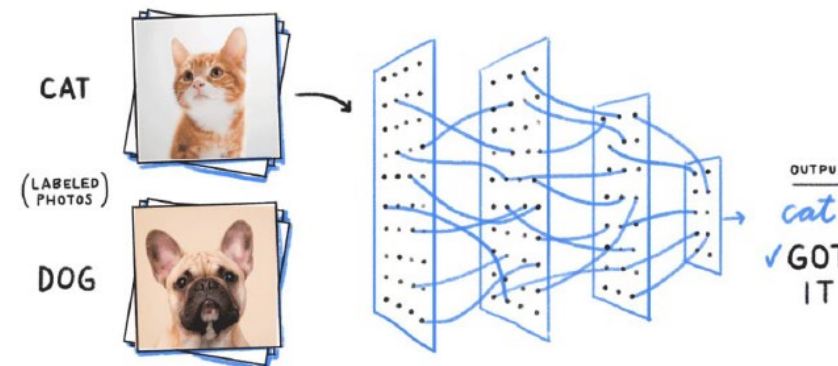
- Development of computer systems
- Performing tasks requiring human intelligence



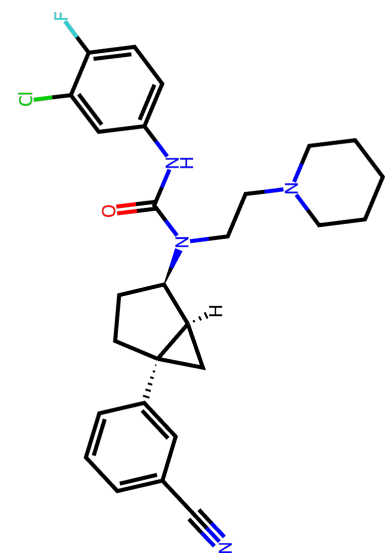
https://link.springer.com/chapter/10.1007/978-981-10-9035-6_82

ML: Learning from examples

DL: Deep neural networks, learning representations



The Path From Molecules to Properties



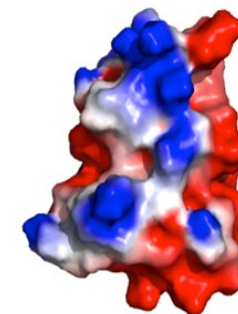
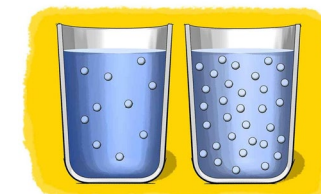
Supervised Learning (SL)

Representation

- Physchem properties
- Morgan FPs
- Adjacency matrix
- Graphs, ...

Model

- RF
- SVM
- FNN/MLP
- GNN



Self-Supervised Learning (SSL)

Tokenization

- Sequences
- Graphs

Foundational Models

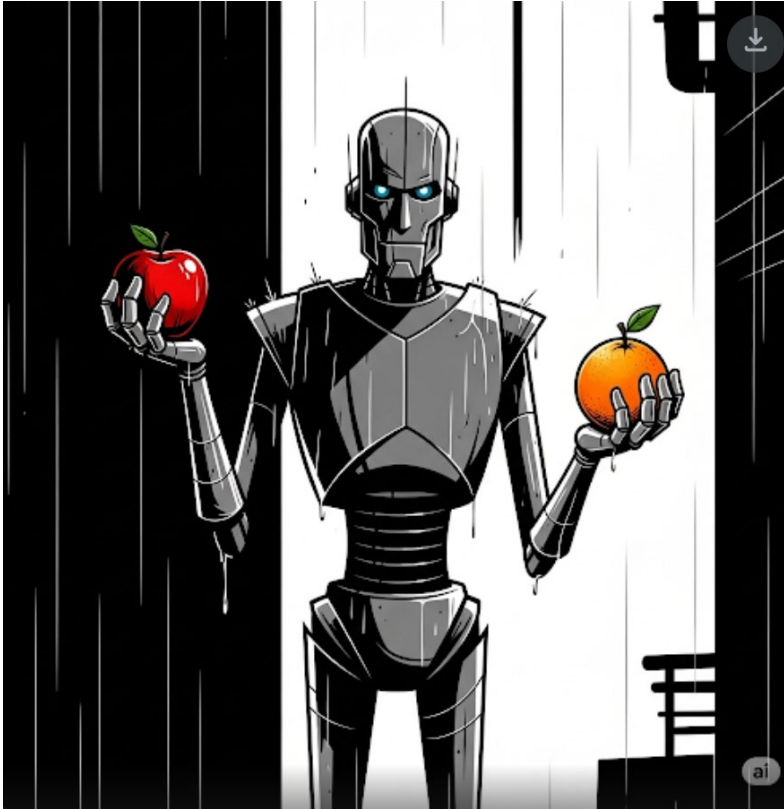
- Transformers
- Autoencoders

Fine-tuning



Generative AI in Still an Evolving Field

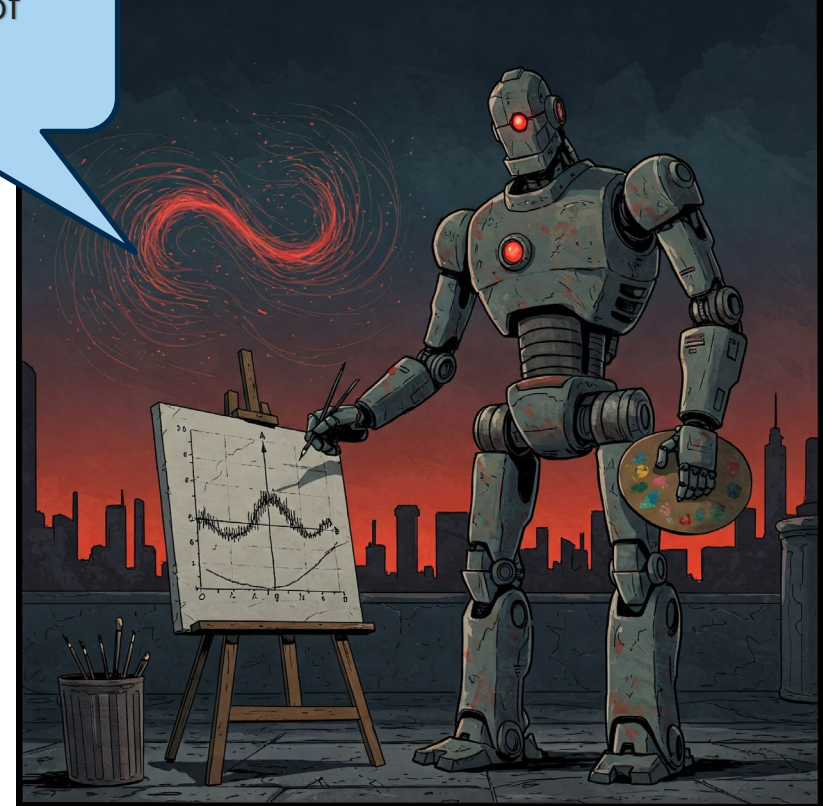
Discriminative AI



Labeling things using examples

The subject of
Session 3

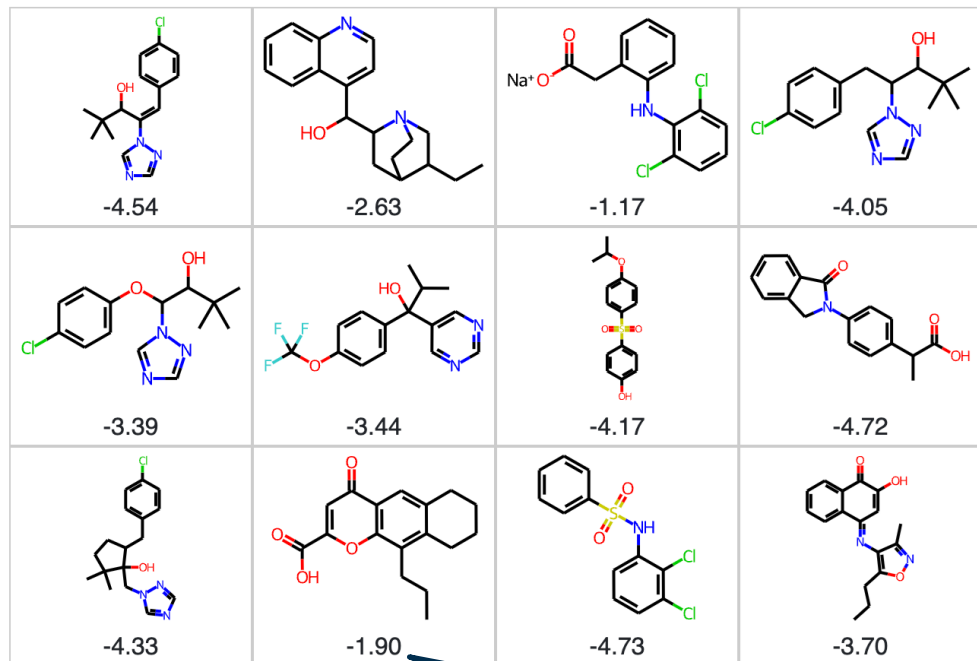
Generative AI



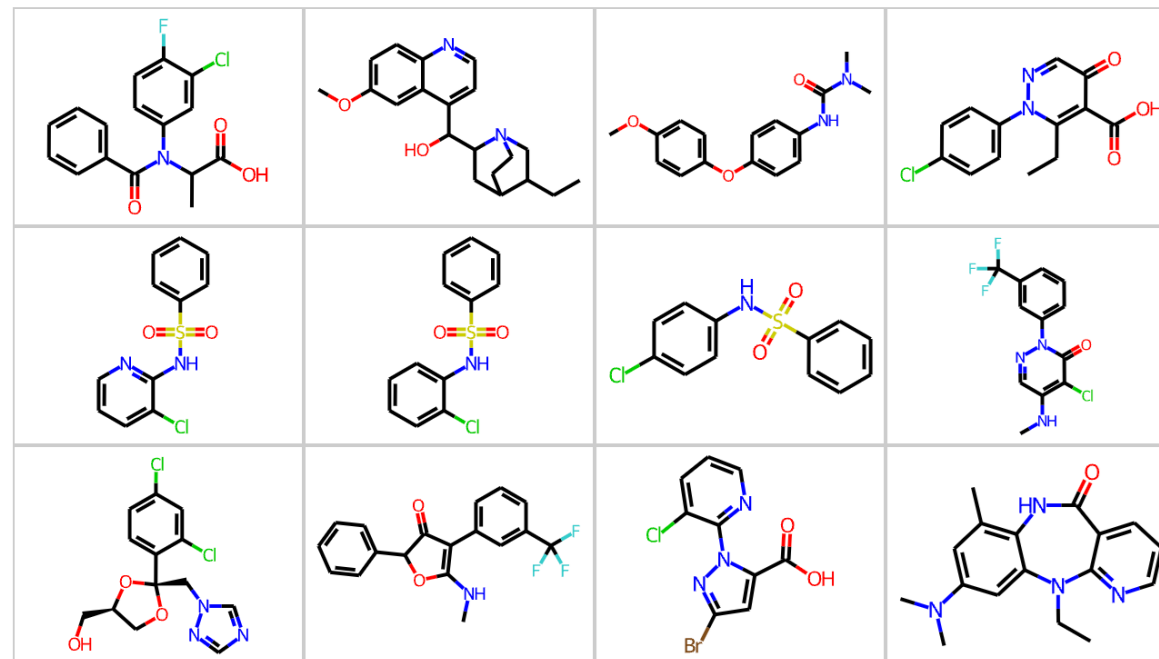
Create new examples from distributions

ML is all About Labeling *Molecules* Based on Examples

Molecules with measured data



Molecules to be predicted

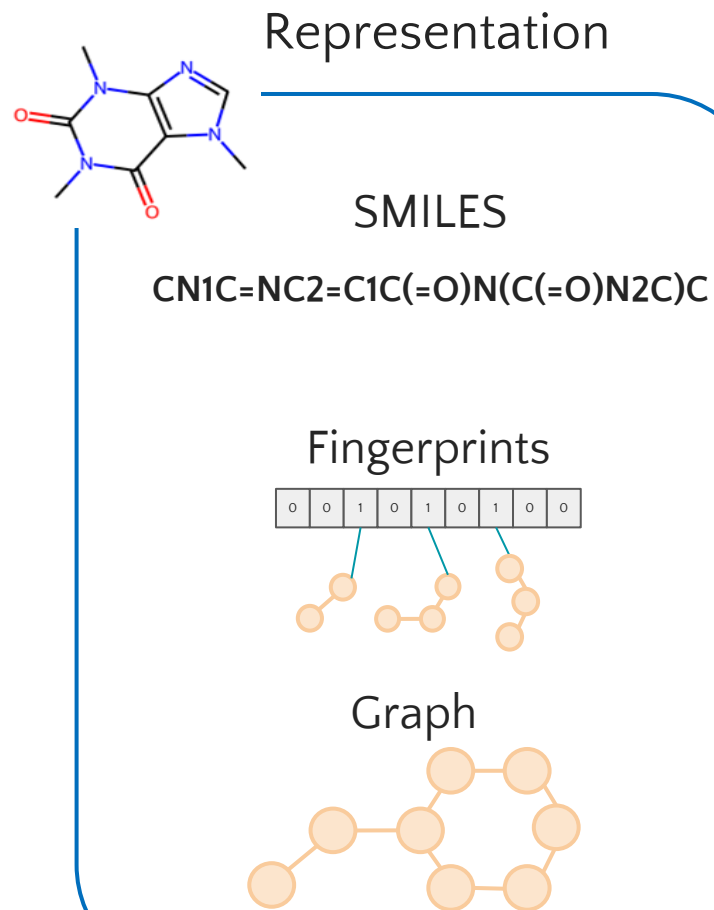


Log10(Molar Aqueous Solubility)

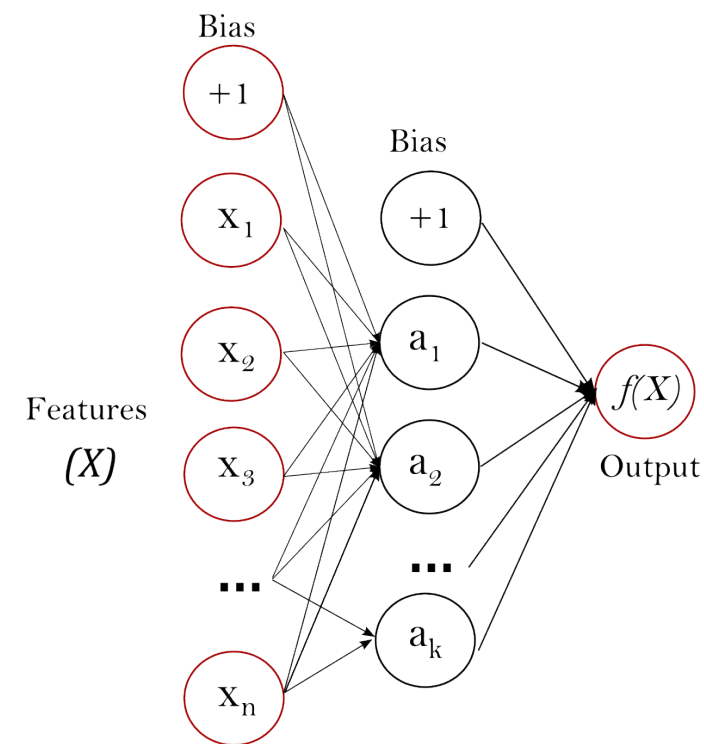
Key Components of Machine Learning in Drug Discovery (or Anything Else)

(Training) Data

0.0009	0.004	0.0008				26.7	4.2		0.232		
0.001	0.004	0.0006				36.2	5.2		0.179		
0.001	0.736	0.0076									
0.001	0.004	0.001				27.5	3.5		0.262		
0.001	0.017	0.0004	<1.0			30.5	3.8		1.4		
0.0015	0.041	0.0415	38.3						12		
0.002	0.02	0.028	385.2			30	2.5		0.583		
0.001	0.0005	0.004	386.7			24.4	3.5		3.1		
0.0006	0.0006	0.0026	326.5	34	5.2	70.4	3.2	496	9.15	0.25	0.291
0.001	0.0145	0.001	4.1			26.2	3.5		1.3		
0.001	3.259	0.0048	5.3						2.2		
0.001	0.007	0.0026	375.0			22.5	3.4		0.395		
0.0016	0.73133	0.0003	6.9	186.3	4.3	72.1	3.8	7.3	16.45	1	7.4
0.001	0.007	0.0008	337.0			26.8	3.9		5		
0.0006	0.0075	0.0008	56.2			5.9	4.3		0.218		
0.001	0.0095	0.002							3.1		
0.0025	16.885	0.121									
0.0028	0.054687	0.0305	42.2	4		27.1	4.5	28.4	2.4	2	0.090
0.0007	0.006	0.004				21.6	36.2		6		
0.001	0.001	0.0003				53.0	5.3		7.5		
0.001	0.207	0.026				0.491	6.2		36		
0.001	0.2715	0.0041							0.007		
0.001	0.3515	0.006							0.363		



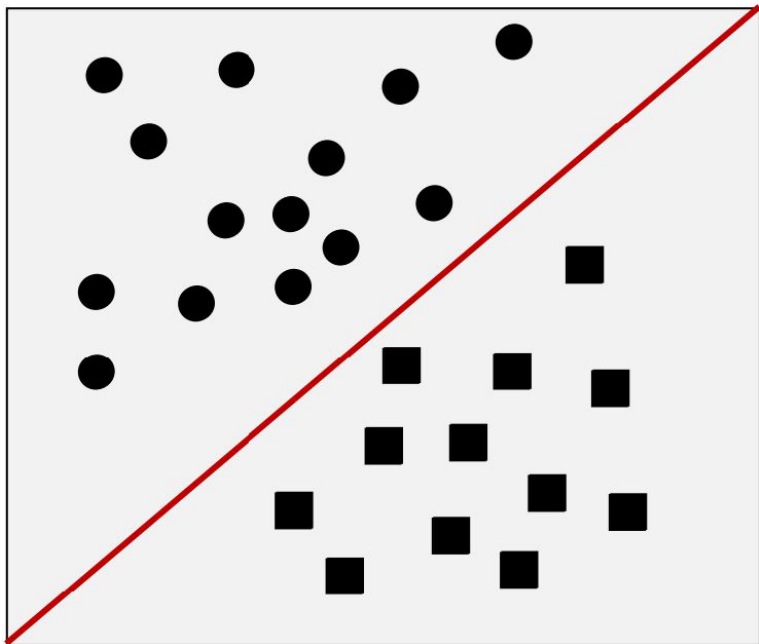
Architecture



Section 1

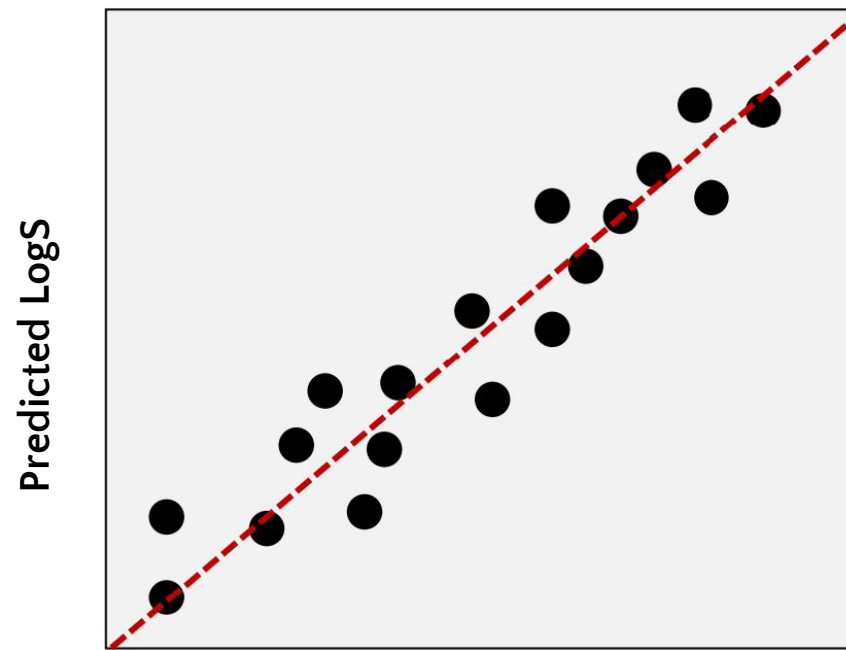
Two Types of ML Models – Classification and Regression

Classification



Predict a category, e.g.
soluble/insoluble

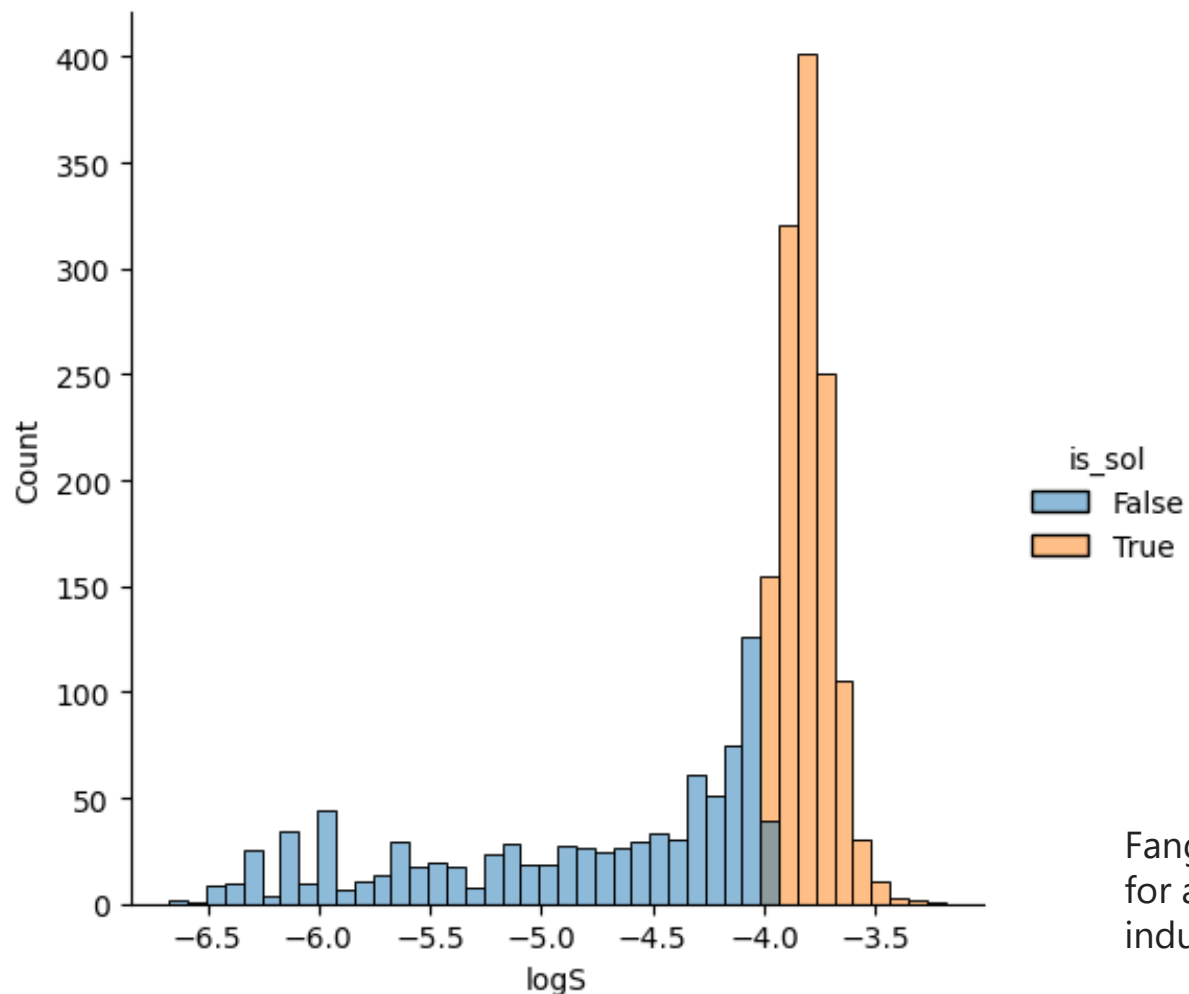
Regression



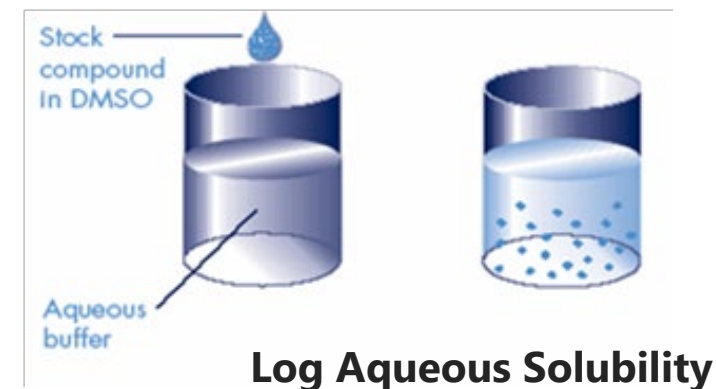
Experimental LogS

Predict a value, e.g. 4.2

Let's Start With a Dataset

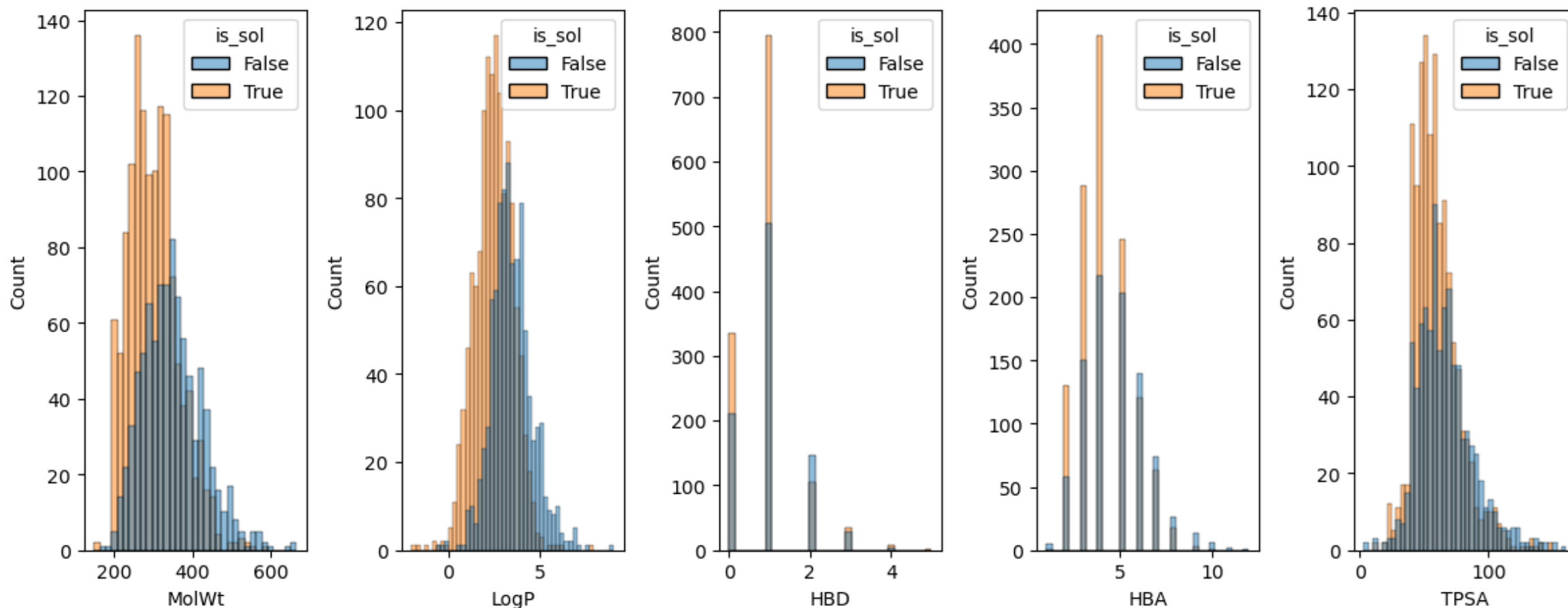


Aqueous solubility by CLND
2173 compounds
Min = 0.23 μ M
Max = 648 μ M



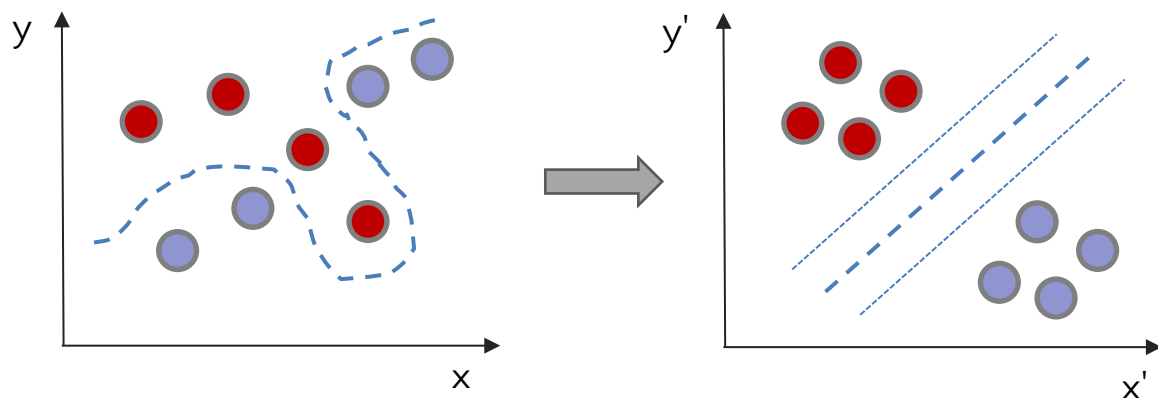
Fang, Cheng, et al. "Prospective validation of machine learning algorithms for absorption, distribution, metabolism, and excretion prediction: An industrial perspective." *JCIM*, 63.11 (2023): 3263-3274.

Which Property Best Separates Soluble vs Insoluble Molecules?



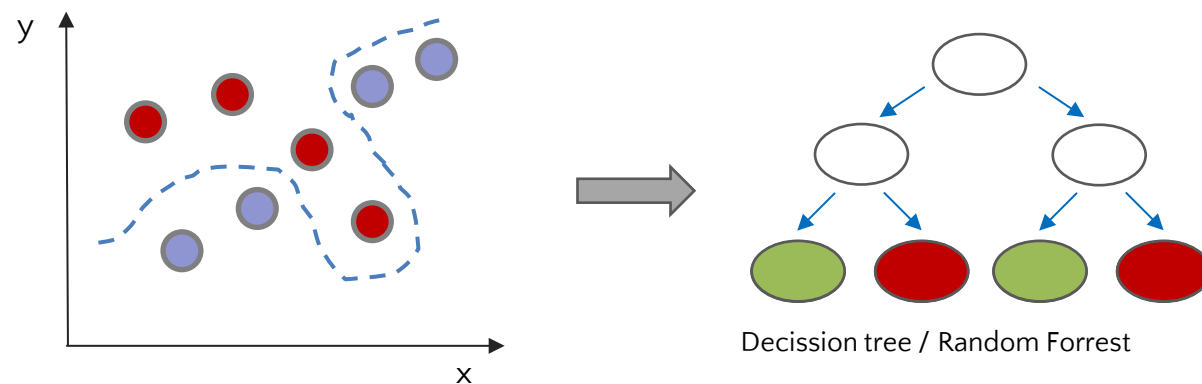
Machine Learning Architectures – SVM & RF

Classification of not linear separable data: Nearest neighbors, Naive Bayes, **SVM**, RF



Support Vector Machine (SVM):

- Transformation of original data to higher dimension via Kernel function
- Hyperplane to separate patterns



Random forest (RF):

- Combines multiple decision trees
- Random subsets of features for diversity
- Majority voting/averaging

Build a Decision Tree – The Gini Index Quantifies the “Purity” of a Split

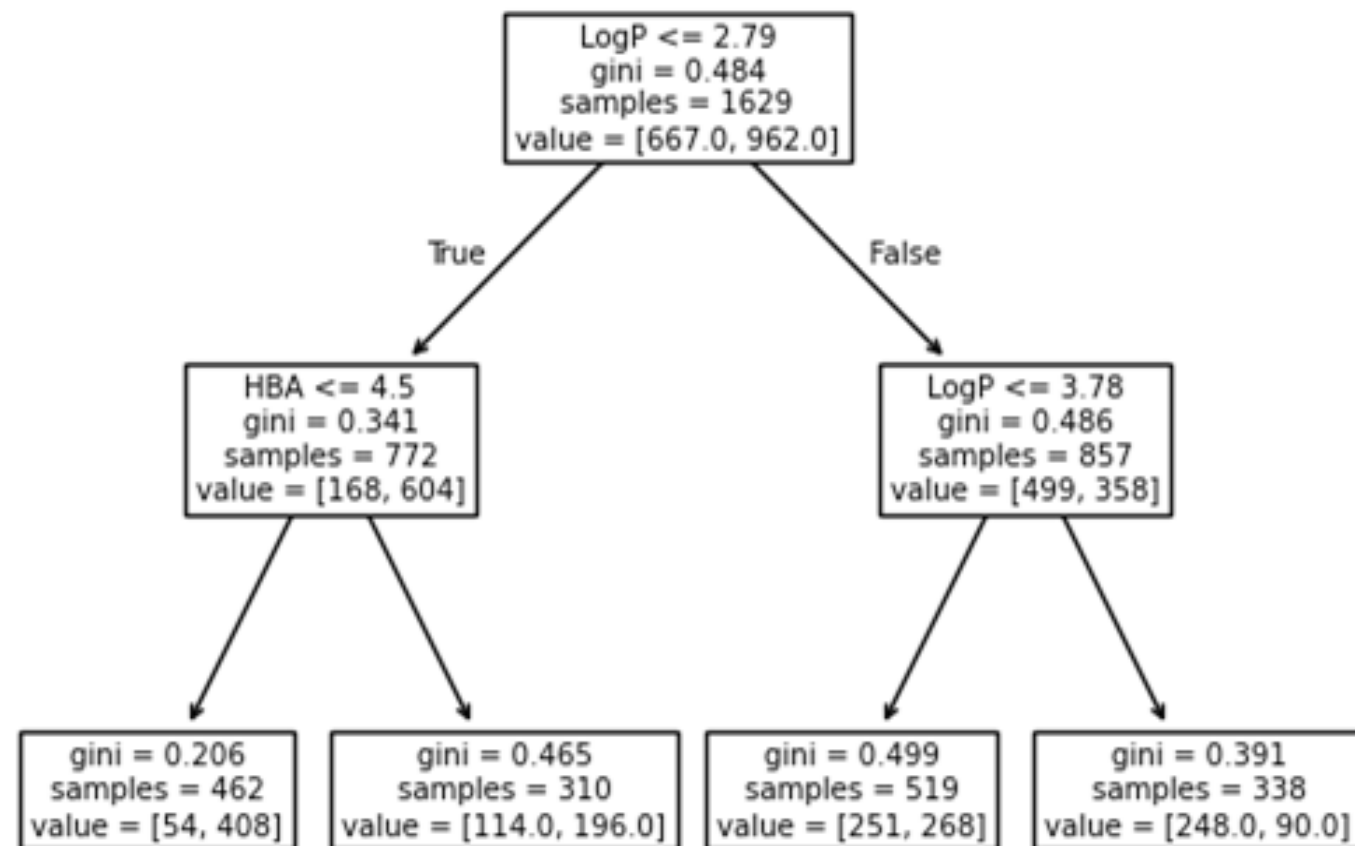
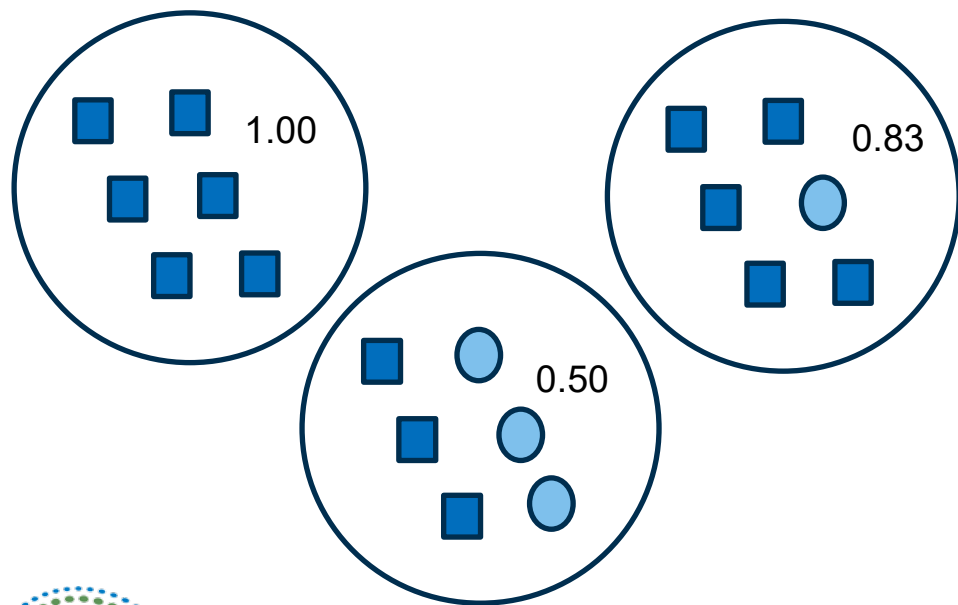
- Binary set-up: How good does this node splits the data into 2

$$\text{Gini Index} = 1 - \sum_{i=1}^n (P_i)^2$$

P_i = proportion of samples belonging to class i

n = number of classes

Binary setting, gini range [0,0.5]



Build a Decision Tree – The Gini Index Quantifies the “Purity” of a Split

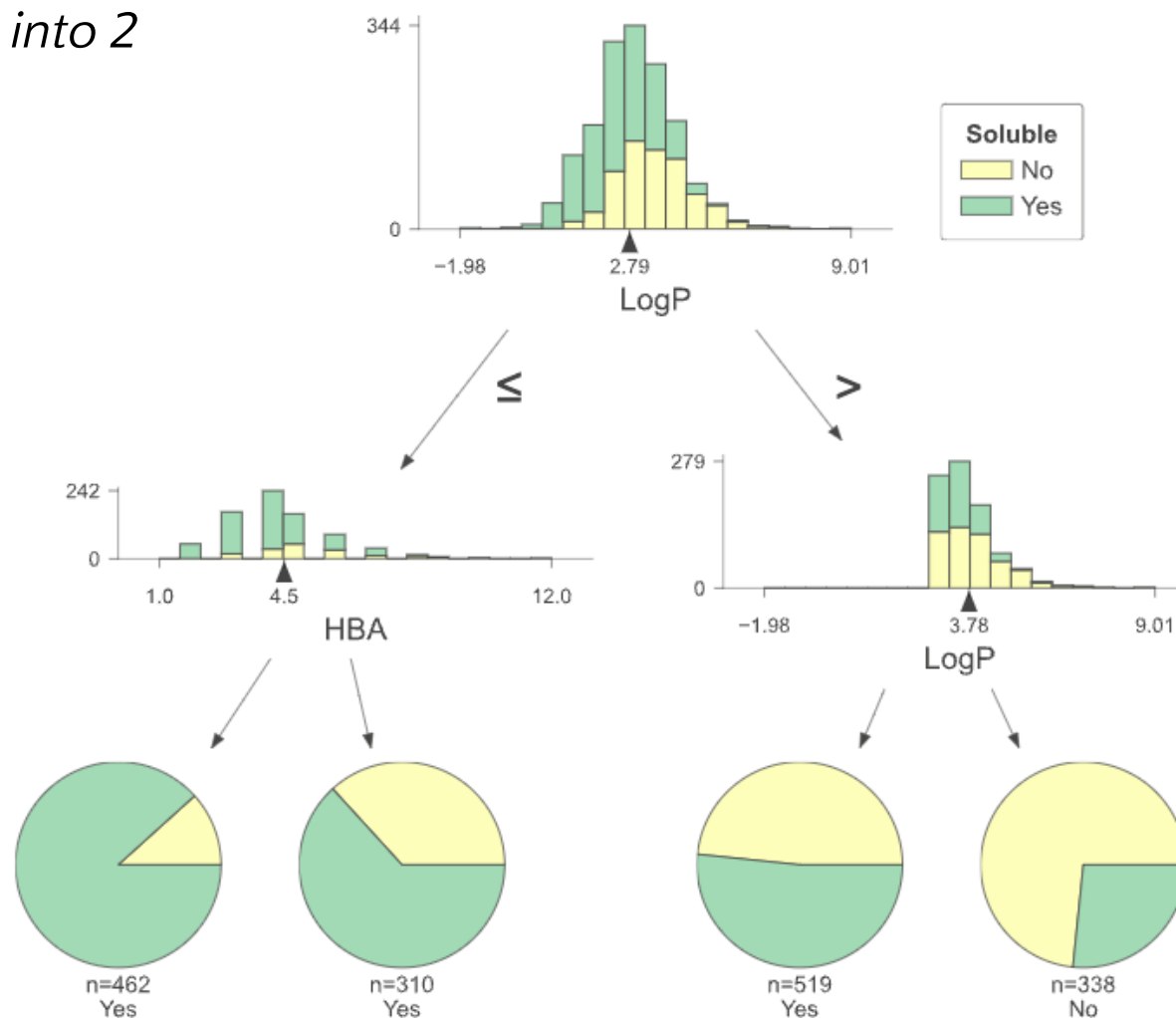
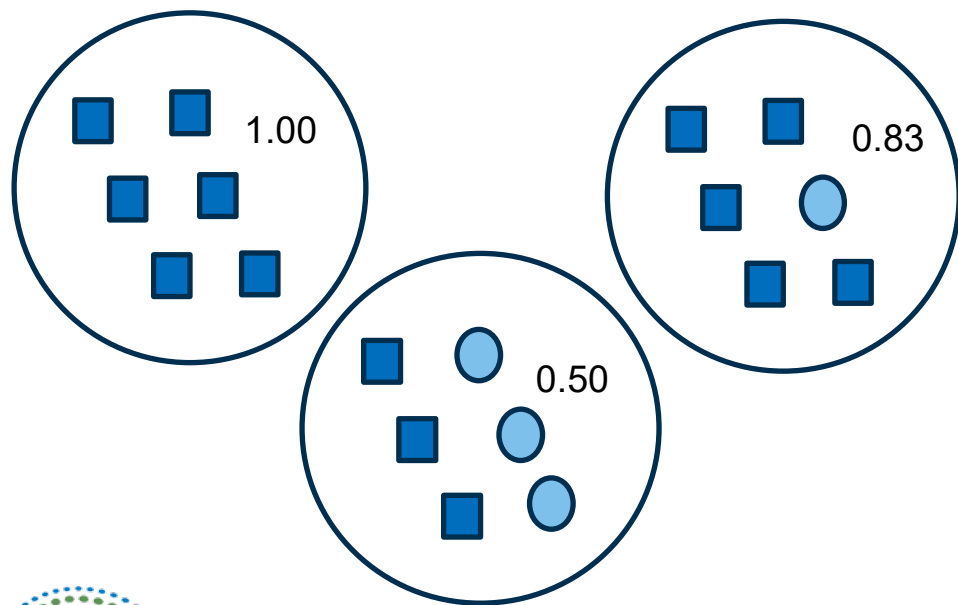
- How good does this node splits the data into 2

$$\text{Gini Index} = 1 - \sum_{i=1}^n (P_i)^2$$

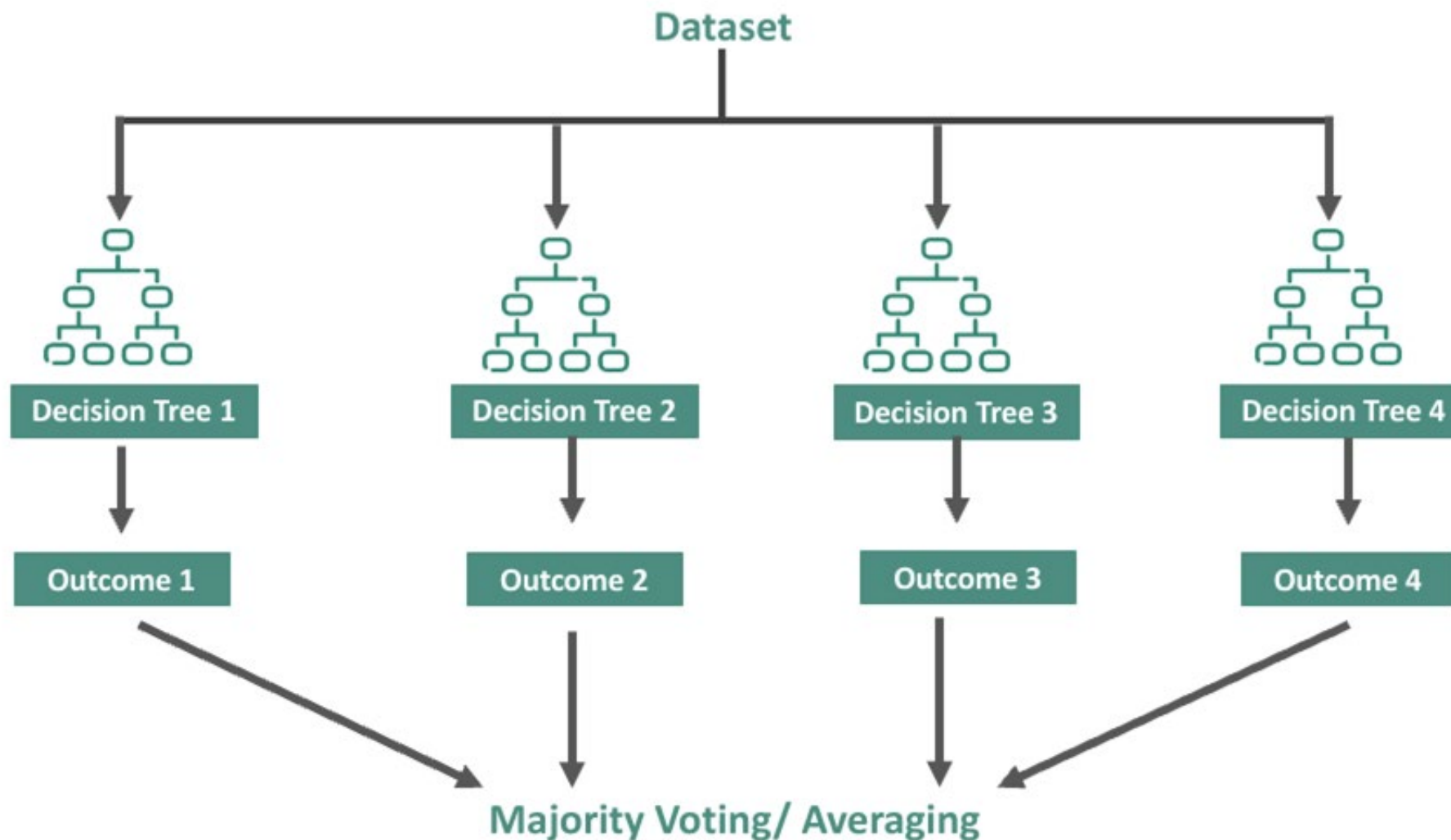
P_i = proportion of samples belonging to class i

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Binary setting, gini range [0,0.5]



Random Forest Uses an Ensemble of Decision Trees

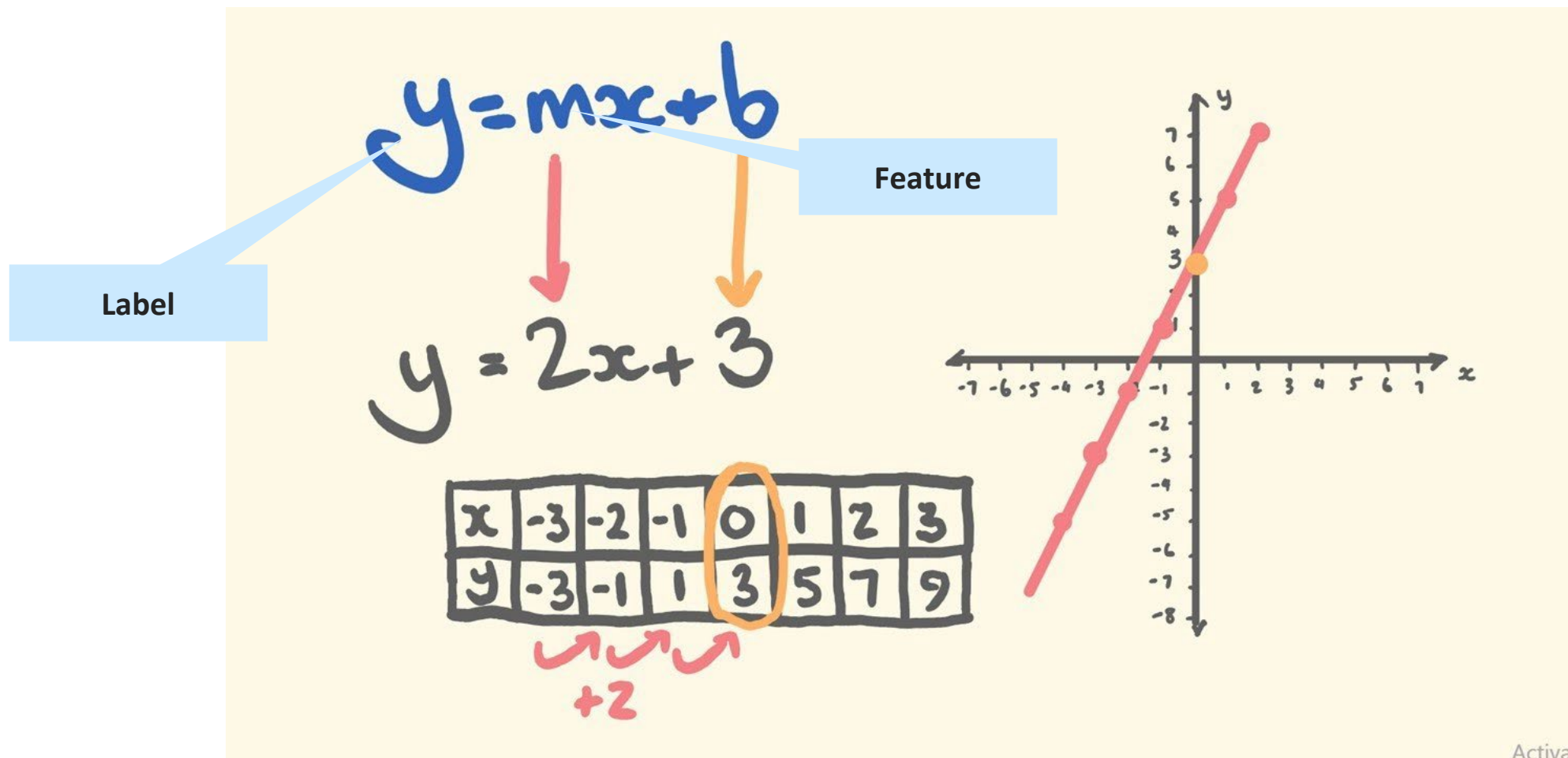


There Are Many Tree Ensemble Methods

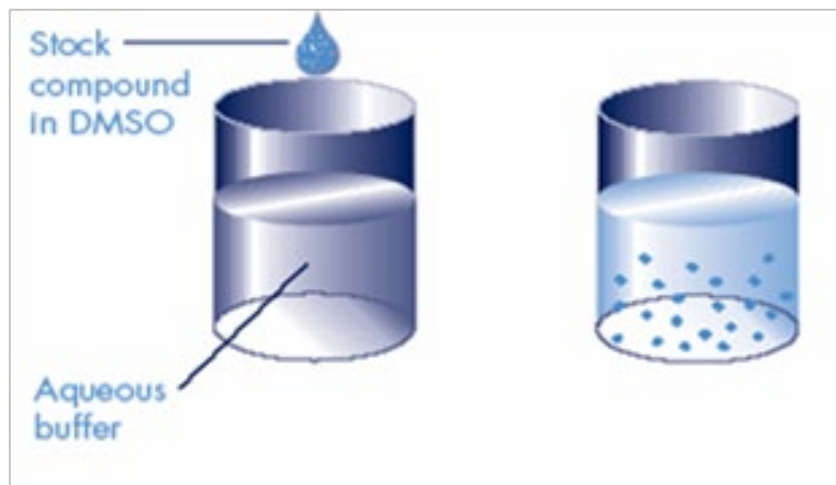


XGBoost

ML Predicts a Set of Labels (y) Based on Features (X)

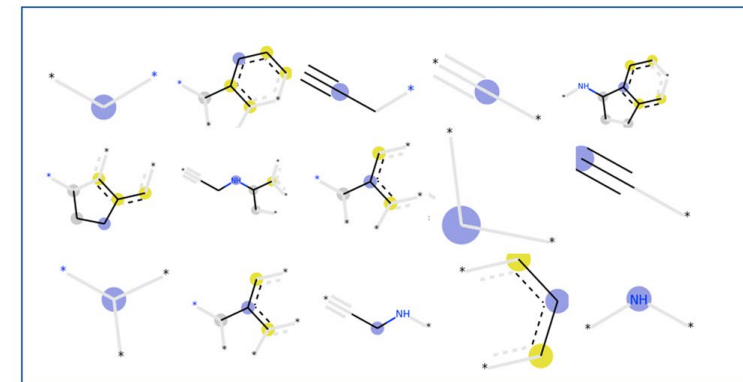
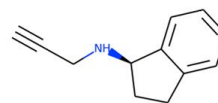


Defining Labels (y) and Features (X) - An Example



y = Log Aqueous Solubility

Label



010110010001000111001000110010101010111

Features

X = A Vector Representing a Molecule

Representation and Encoding of Molecules

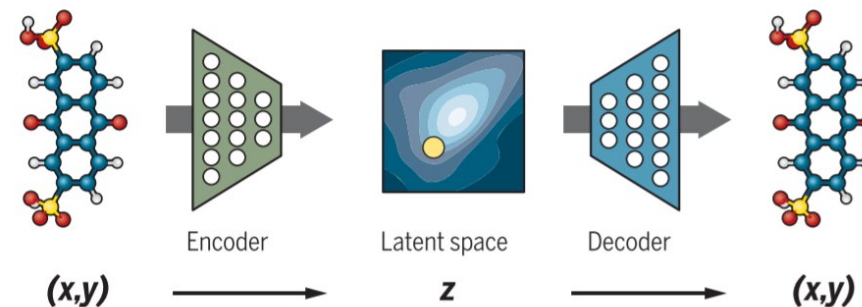
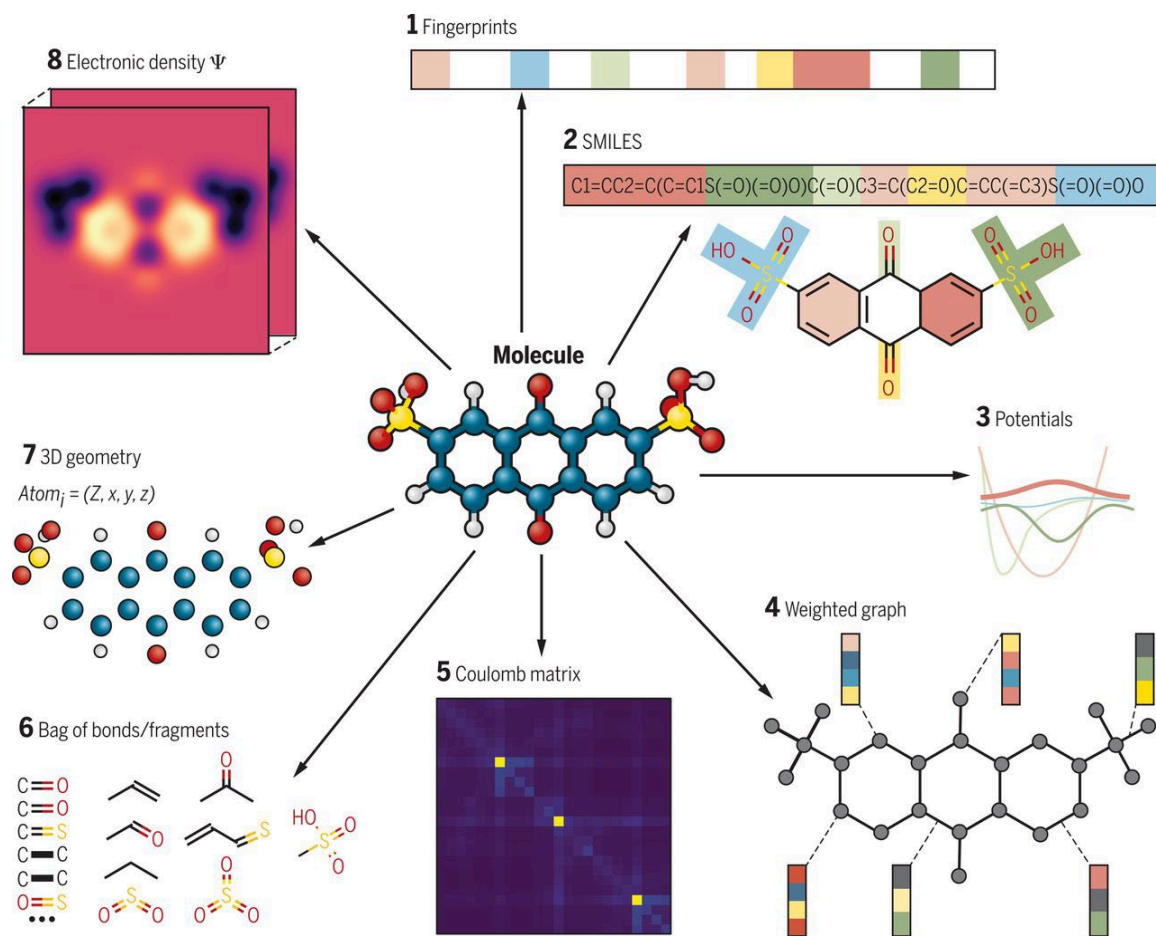
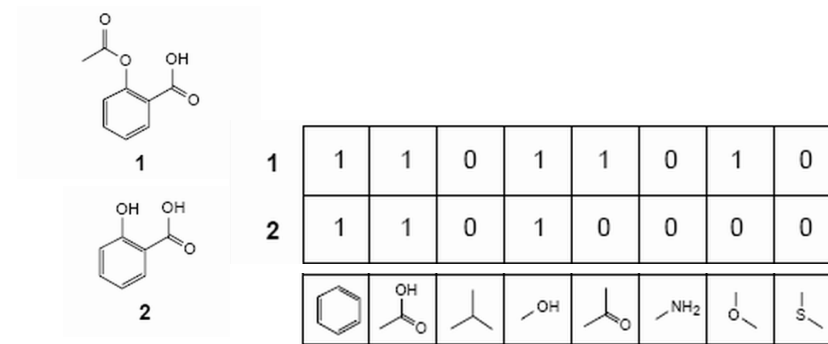


Figure taken from: Sanchez-Lengeling et al., Science 361, 360–365 (2018)

Exemplary Tabular Fingerprint Designs

- Fragment-based fingerprints

- Report presence or absence of structural fragments
- **MACCS** (166 bit)



- Topological atom environments

- Generate strings of layered atom-centered connectivity patterns (circular FPs)
- **ECFP/Morgan** or Molprint2D (varying numbers of strings)



- 2D/3D pharmacophore fingerprints

- Record potential pharmacophoric features (up to millions of bits)



Representation and Encoding of Molecules

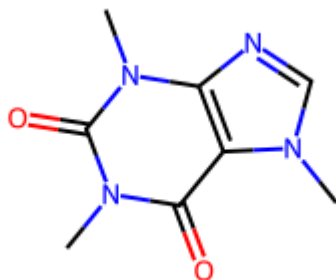
```
from rdkit import Chem
from rdkit.Chem import Draw
from rdkit.Chem import AllChem

# Input SMILES string
# Example: caffeine
smiles = "CN1C=NC2=C1C(=O)N(C(=O)N2C)C"

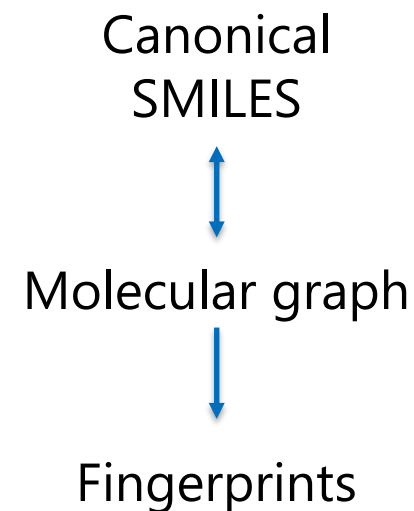
# Convert SMILES to molecule
mol = Chem.MolFromSmiles(smiles)

# Generate Morgan fingerprint
fingerprint = AllChem.GetMorganFingerprintAsBitVect(mol, radius=2, nBits=1024)

# Draw the molecule
mol
```



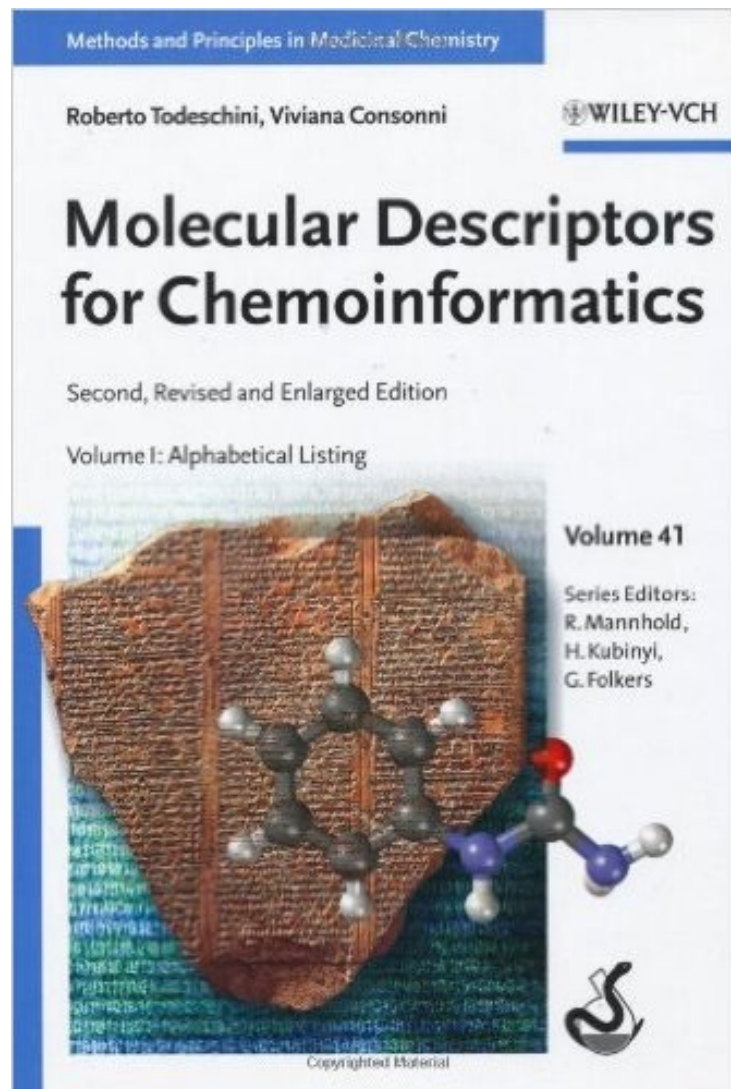
Working with molecules



Open-Source Cheminformatics
and Machine Learning

<https://www.rdkit.org/>

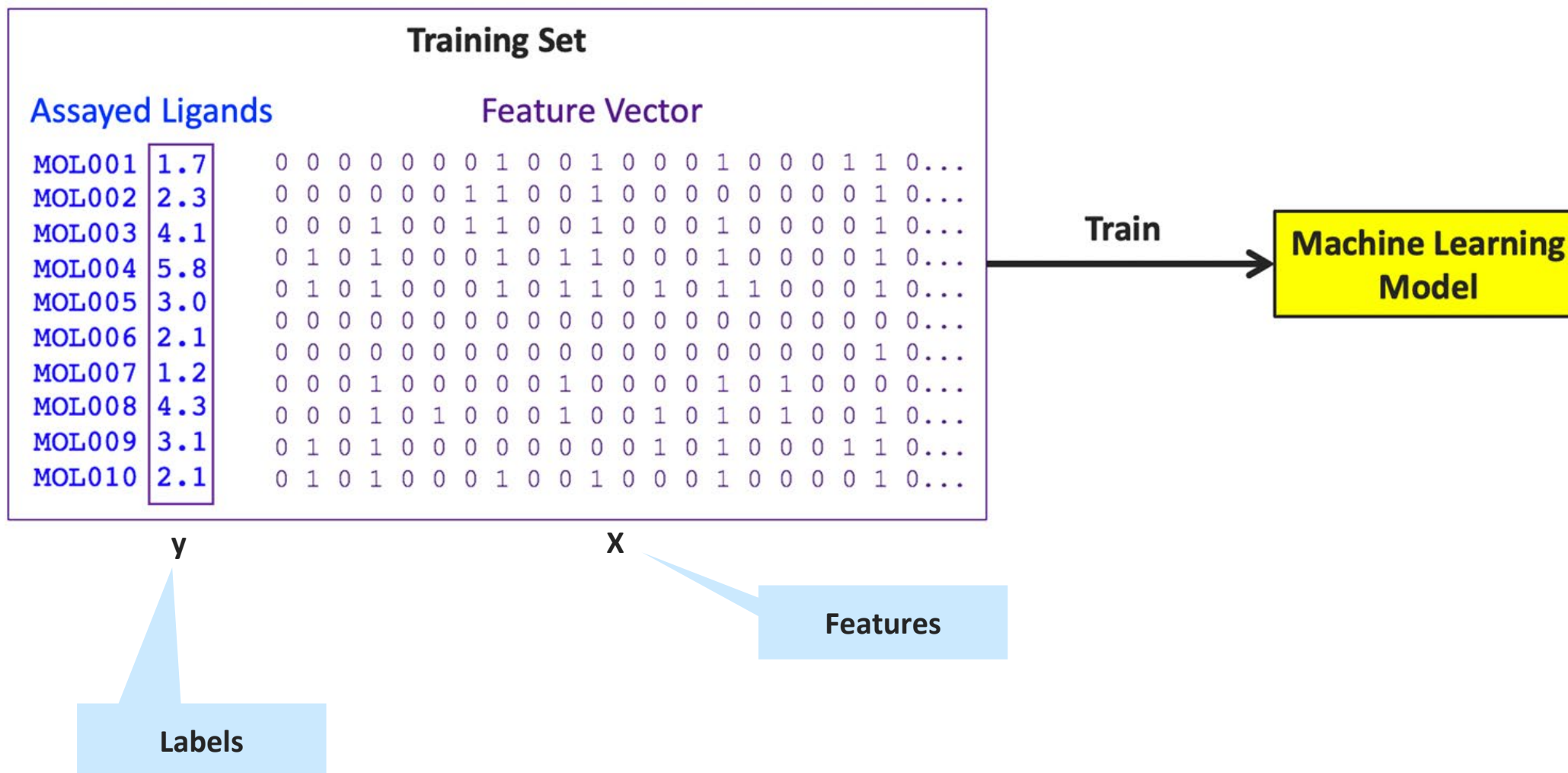
There are A LOT of ways to do this



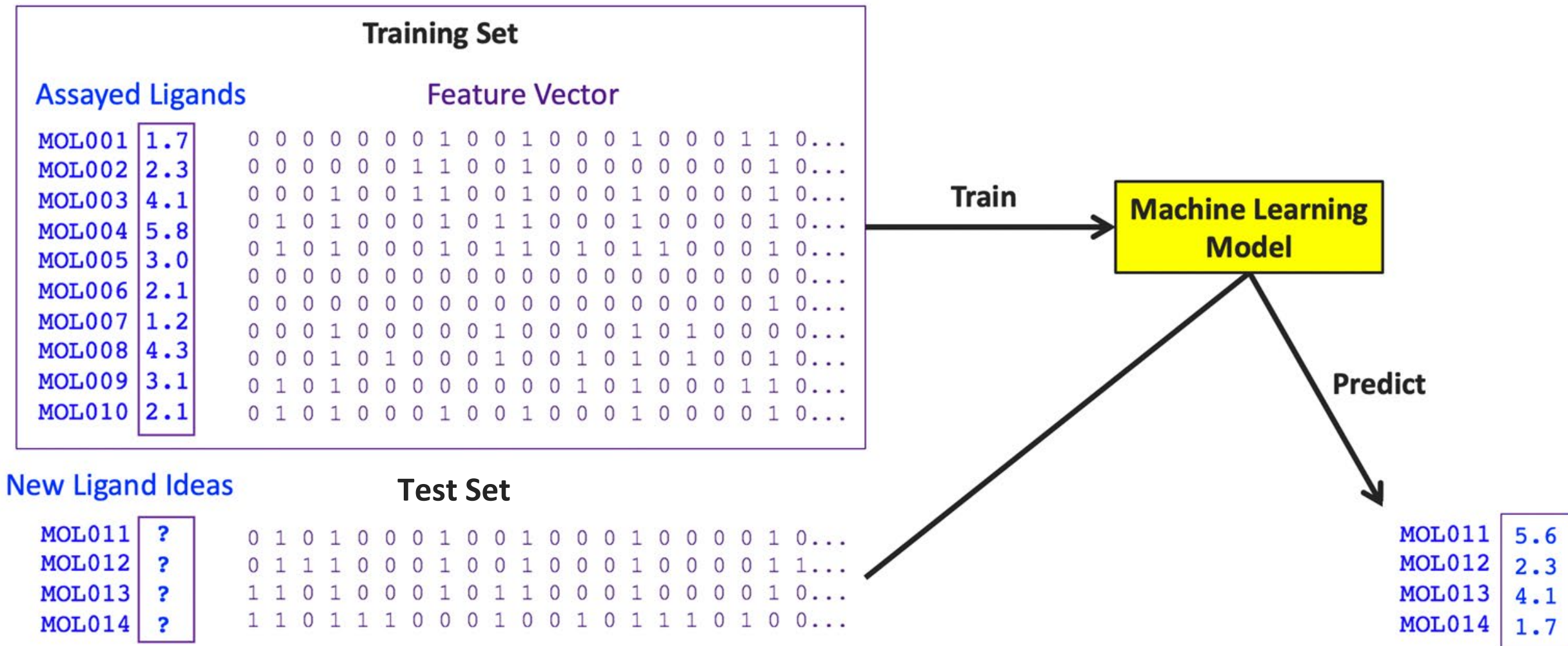
2 Volumes
6,000 references from 450 journals

DRAGON 7 has 5,270 descriptors z

Training a Machine Learning Model



Making Predictions With a Machine Learning Model



Training a Model that Generalizes Well to New Data

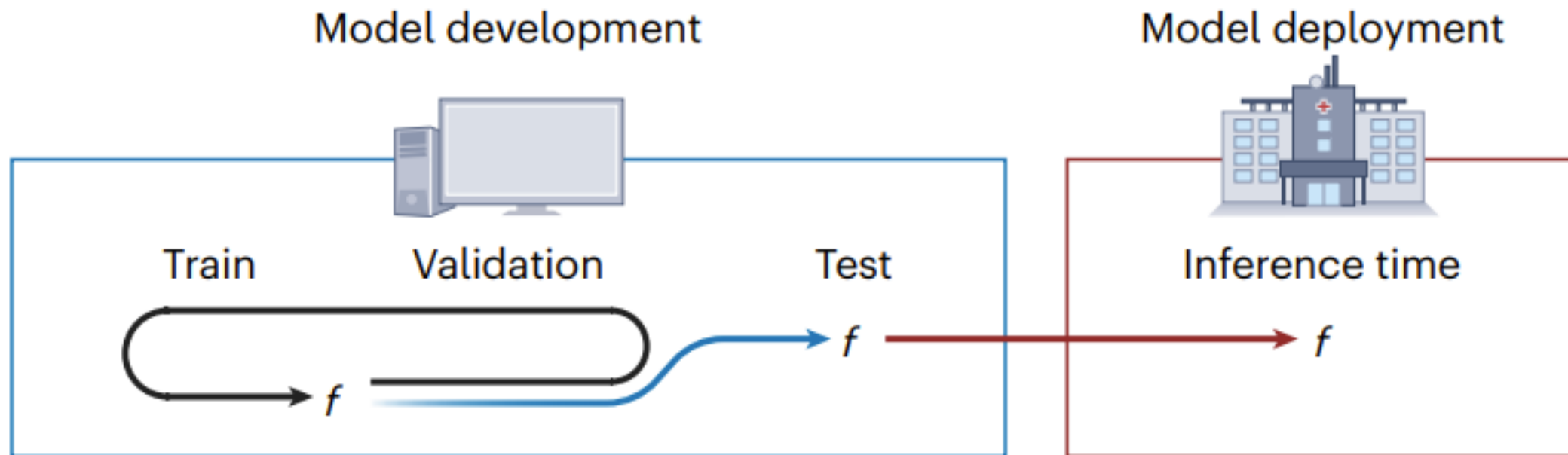


Fig. taken from <https://doi.org/10.1038/s41592-024-02362-y>

Prediction Performance of an Aqueous Solubility Model

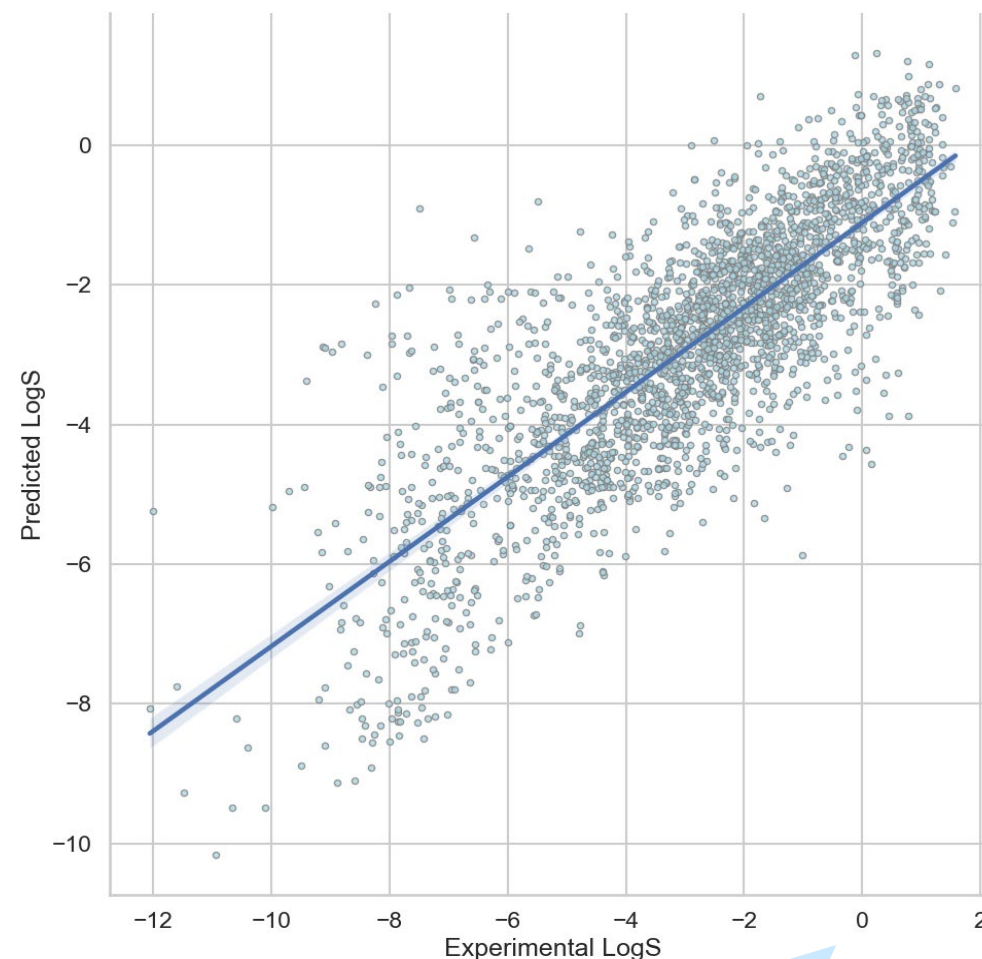
Prediction performance of an aqueous solubility model

- RMSE (root-mean-squared error)
- Average magnitude of the errors between predicted values and actual values

$$\text{RMSE} = \sqrt{\frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2}$$

- n : Number of data points.
- y_i : Actual value of the target.
- $\hat{y}_i$: Predicted value of the target.

➤ More in second part!



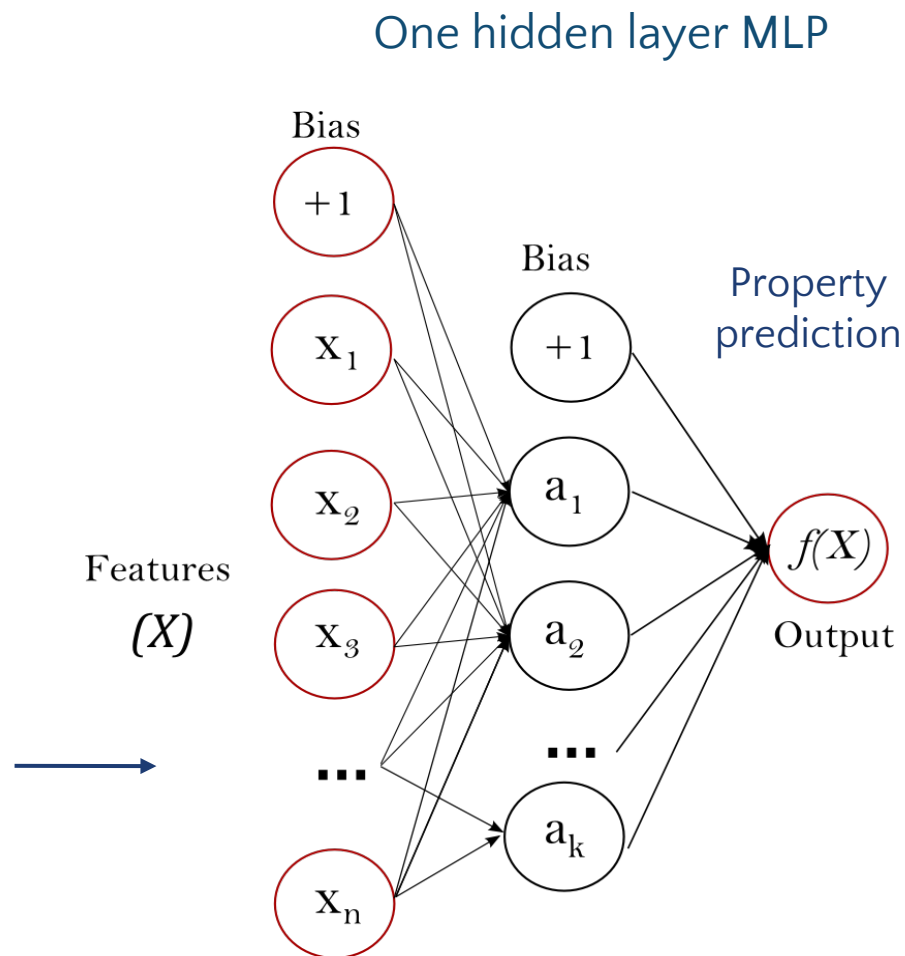
Dynamic range is very large, does not represent a typical use case

Neural Networks in a Nutshell

Deep Learning Architectures – Neural Networks

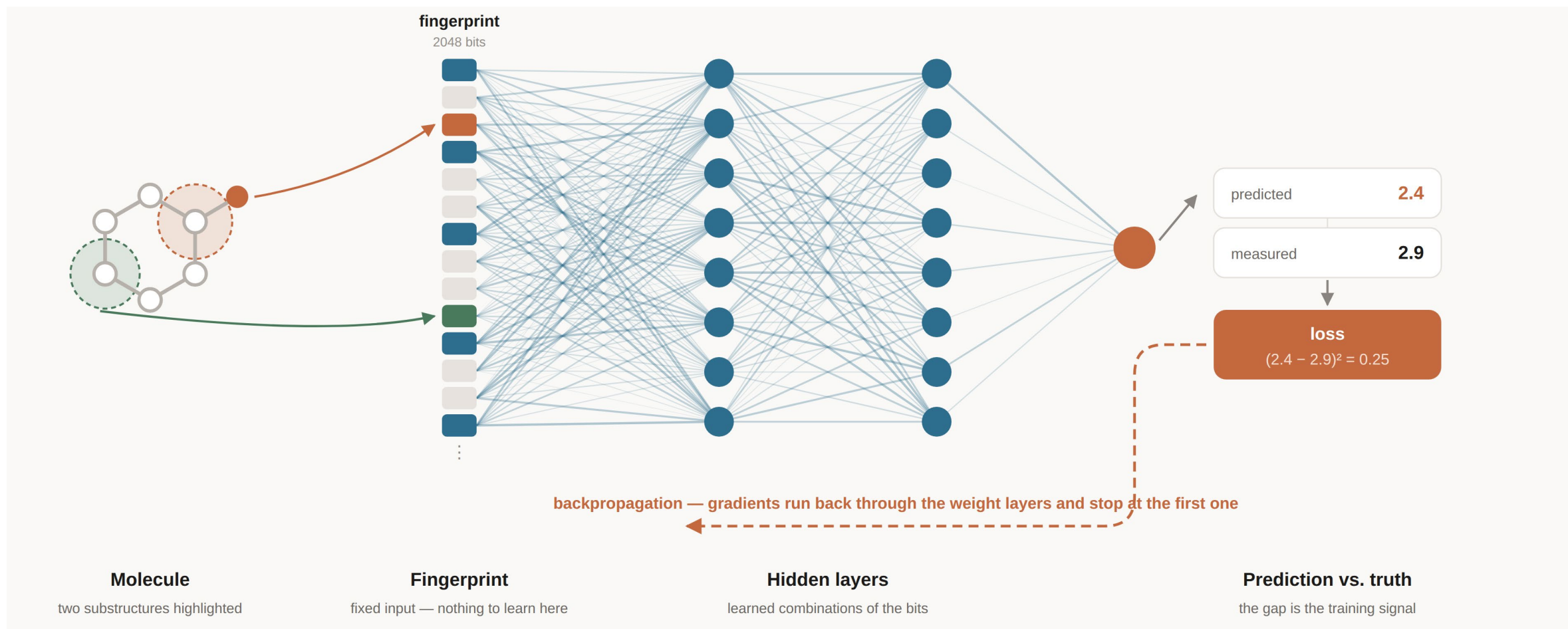
Extremely successful in large-data settings

- Can learn complex patterns
- Architecture depends on input
 - Molecular fingerprint: Multi-layer-perceptron (MLP) / Feed-forward Neural Network (FFNN) / Fully-connected NN (FCNN)
 - Molecular graph: Graph Neural Network (GNN)
 - SMILES: Recurrent NN (RNN)/transformer



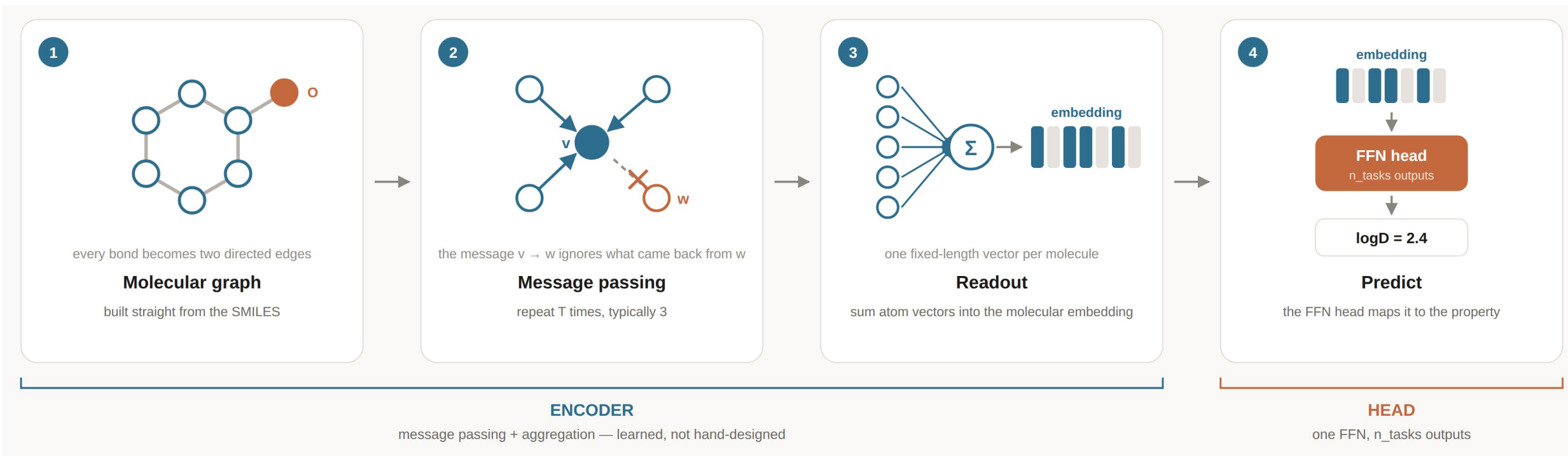
A Deep Neural Network for Molecular Property Prediction

- Forward pass: bits in, prediction out
- Backward pass: the loss tells every weight which way to move.



How a Chemprop Message-Passing Network Works

- Bonds carry the messages
- Atoms collect them
- One vector comes out the end



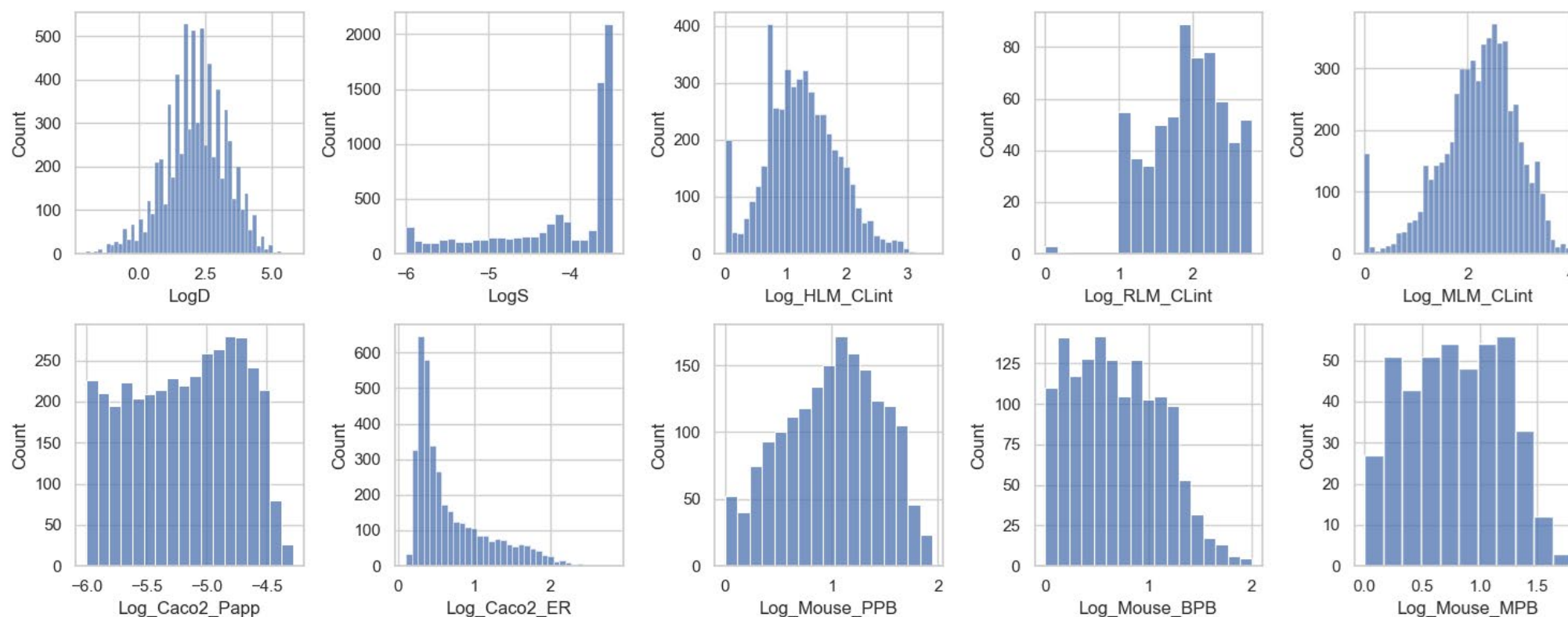
The OpenADMET + ExpansionRx Blind Challenge

10 ADMET Assays

5,326 compound
training set

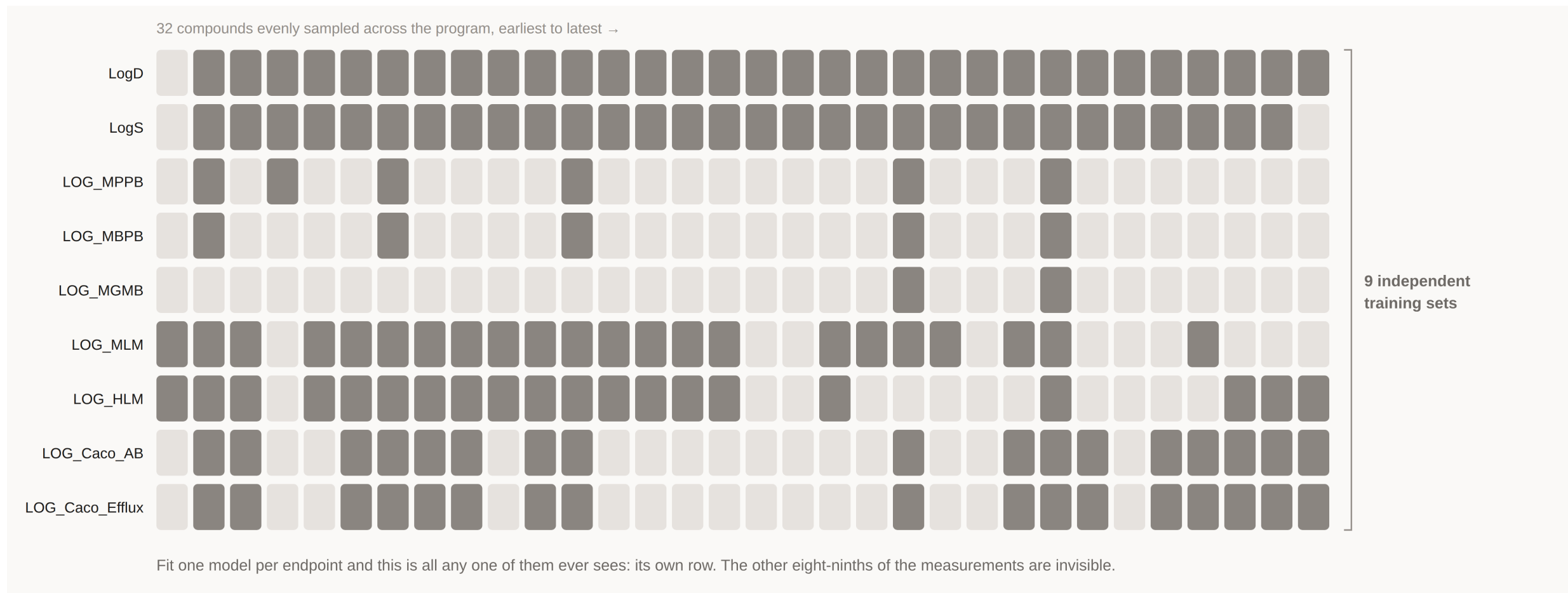
2,282 compound
blinded test set

Time split

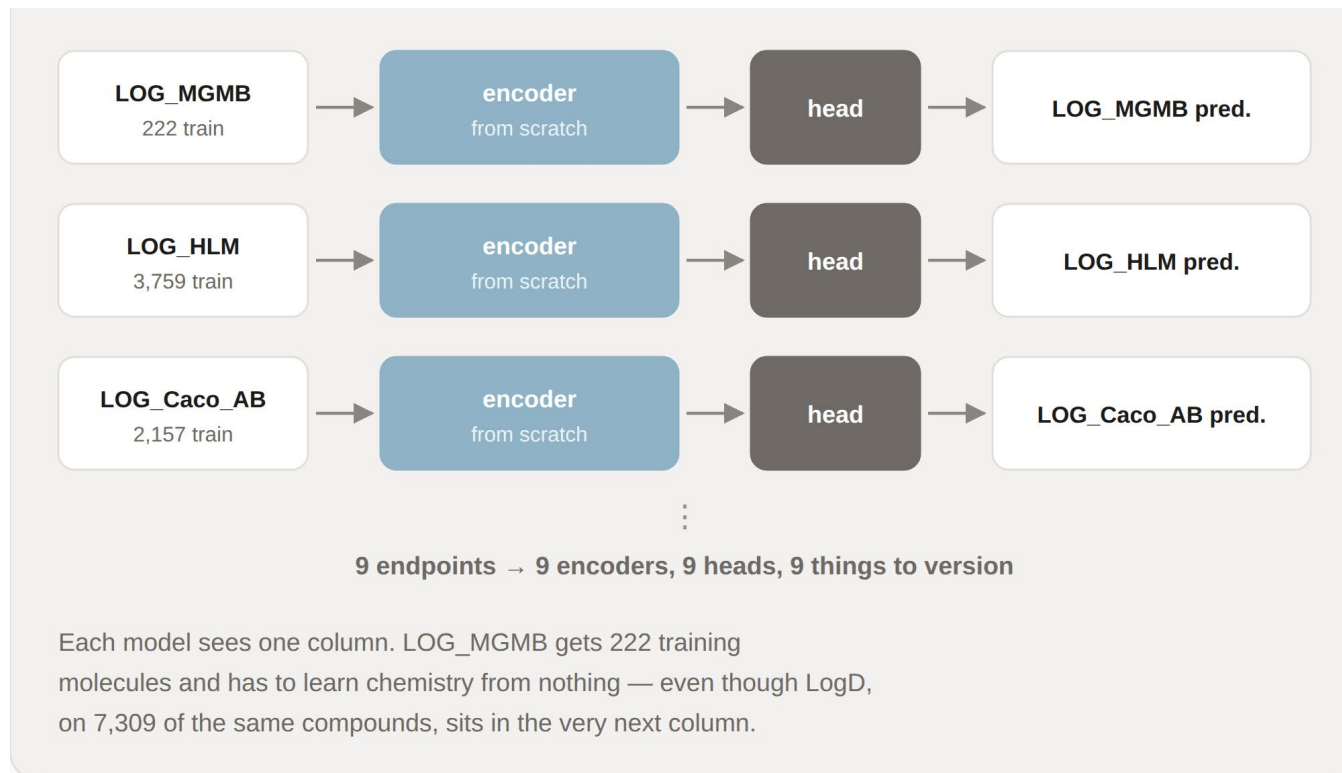


Nine Endpoints, Nine Separate Problems

- The single task view



Single Task – Nine Independent Models

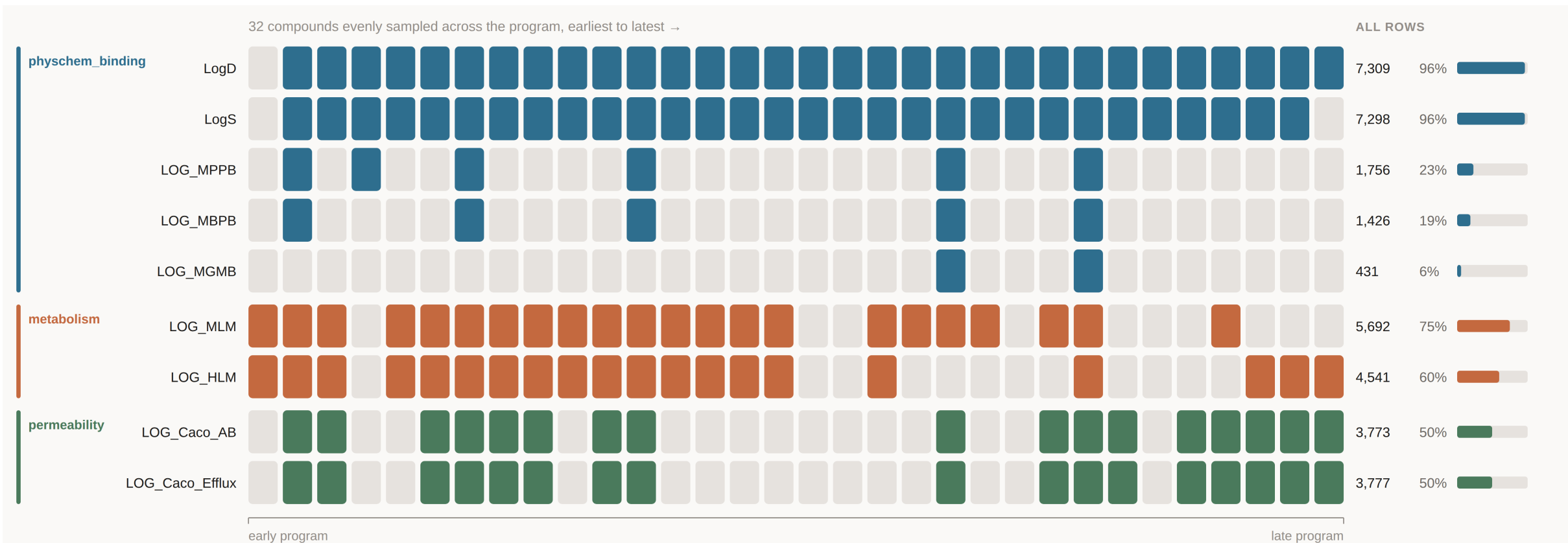


What it costs

- Each model sees only its own column of labels
- LOG_MGMB trains on 222 molecules, from scratch
- Nine encoders, nine heads, nine things to version
- No route for a dense endpoint to help a sparse one

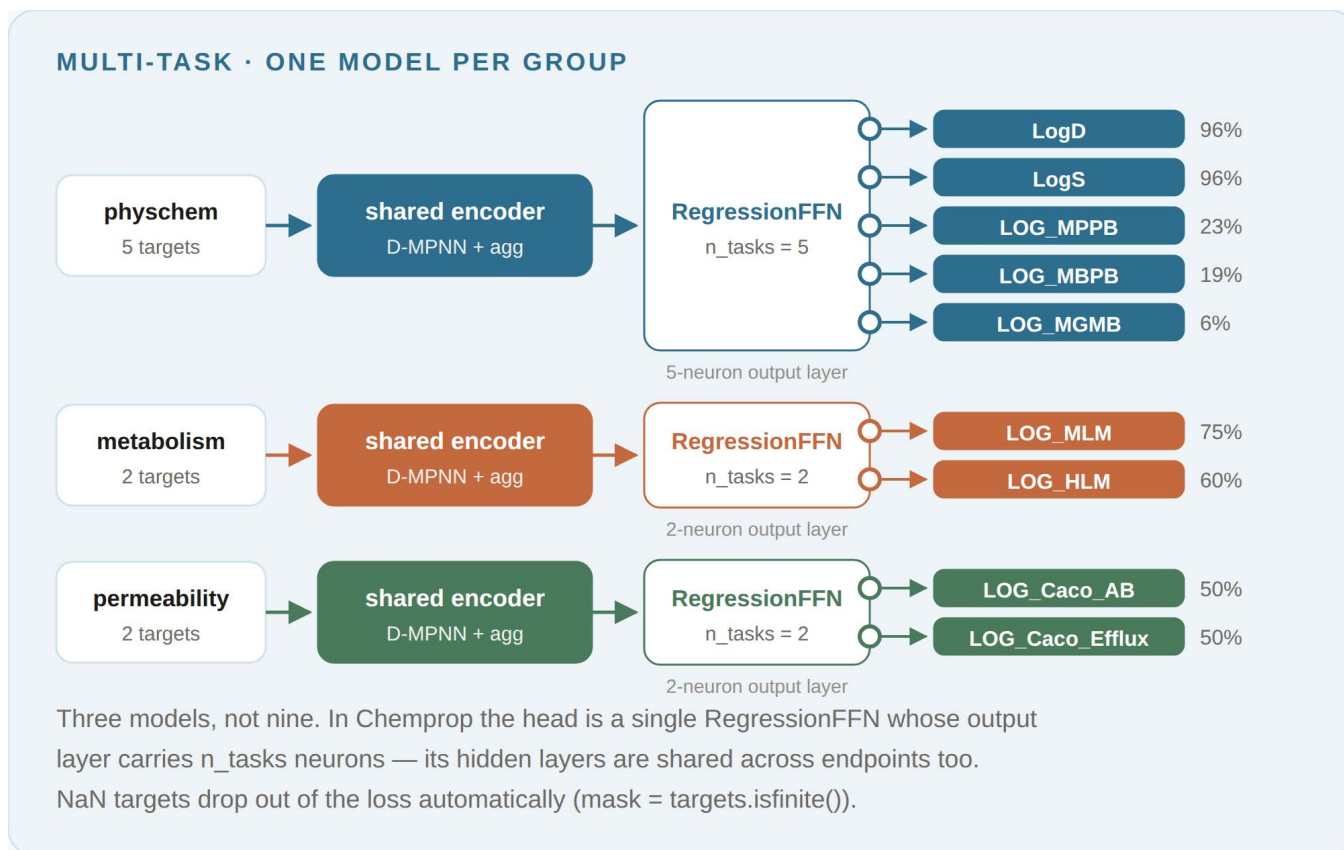
The Same Measurements, Read as One Matrix

- The multi task view



Exactly the same filled cells. Reading them as one matrix is what lets LOG_MGMB — 431 compounds, 5.7% coverage — borrow a representation built from the 7,309 LogD and 7,298 LogS measurements above it. The expensive binding assays only switch on partway through the program, while LOG_HLM tails off.

One Model per Group

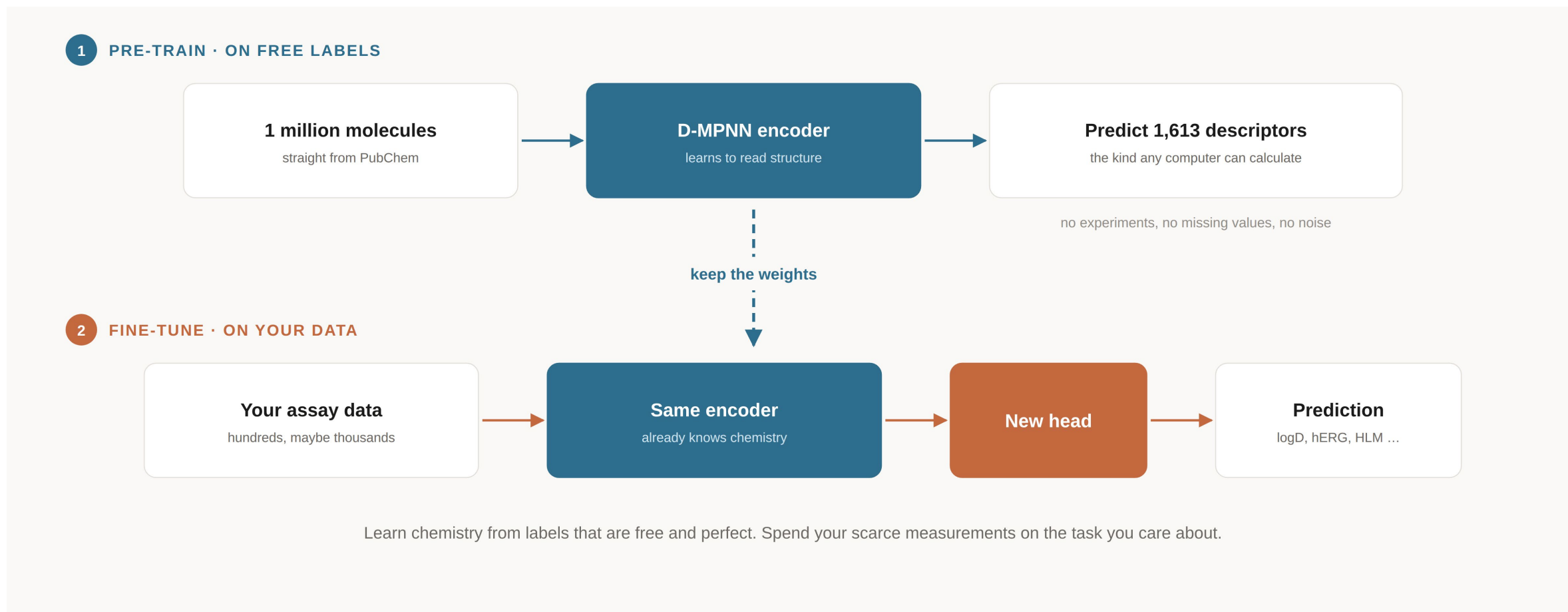


How Chemprop does it

- One RegressionFFN per group, with an n_tasks-neuron output layer
- Hidden layers shared across the endpoints, not just the encoder
- NaN targets drop out of the loss: mask = targets.isfinite()
- task_weights rebalances a 431-compound task against 7,309

CheMeleon: a Descriptor-Based Foundation Model

- Pre-train on labels a computer can generate
- Fine-tune on the assay you care about



How to Find Pat

<https://pwalters.github.io>

Practical Cheminformatics

Publications

Tutorials

Blog

Videos

Resources



Pat Walters

Cheminformatics, ML

📍 Cambridge, MA

✉ Email

🔍 Google Scholar

🐙 Github

🦋 Bluesky

🌐 LinkedIn

✂ X (formerly Twitter)

Pat Walters is Chief Scientist at [OpenADMET](#), an open science initiative that combines insights from high-throughput experimentation, structural biology, and machine learning to improve the prediction of drug absorption, metabolism, excretion, and toxicology. Before his current role, Pat spent thirty years in leadership positions at Relay Therapeutics and Vertex Pharmaceuticals. Pat is the 2023 recipient of the [Herman Skolnik Award](#) for Chemical Information Science from the American Chemical Society. He is a member of the editorial advisory boards for the Journal of Chemical Information and Modeling and Artificial Intelligence in the Life Sciences, and previously held a similar role with the Journal of Medicinal Chemistry. Pat is co-author of the book “[Deep Learning for the Life Sciences](#)”, published in 2019 by O'Reilly and Associates. He received his Ph.D. in Organic Chemistry from the University of Arizona where he studied the application of artificial intelligence in conformational analysis. Prior to obtaining his Ph.D., Pat worked at Varian Instruments as both a chemist and a software developer. He received his B.S. in Chemistry from the University of California, Santa Barbara.



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Thank you for your attention!